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REVIEW

Functional insight into multi-omics-based interventions for climatic resilience in sorghum (*Sorghum bicolor*): a nutritionally rich cereal crop

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Abstract

Main conclusion **The article highlights omics-based interventions in sorghum to combat food and nutritional scarcity in the future.**

Abstract Sorghum with its unique ability to thrive in adverse conditions, has become a tremendous highly nutritive, and multipurpose cereal crop. It is resistant to various types of climatic stressors which will pave its way to a future food crop. Multi-omics refers to the comprehensive study of an organism at multiple molecular levels, including genomics, transcriptomics, proteomics, and metabolomics. Genomic studies have provided insights into the genetic diversity of sorghum and led to the development of genetically improved sorghum. Transcriptomics involves analysing the gene expression patterns in sorghum under various conditions. This knowledge is vital for developing crop varieties with enhanced stress tolerance. Proteomics enables the identification and quantification of the proteins present in sorghum. This approach helps in understanding the functional roles of specific proteins in response to stress and provides insights into metabolic pathways that contribute to resilience and grain production. Metabolomics studies the small molecules, or metabolites, produced by sorghum, provides information about the metabolic pathways that are activated or modified in response to environmental stress. This knowledge can be used to engineer sorghum varieties with improved metabolic efficiency, ultimately leading to better crop yields. In this review, we have focused on various multi-omics approaches, gene expression analysis, and different pathways for the improvement of Sorghum. Applying omics approaches to sorghum research allows for a holistic understanding of its genome function. This knowledge is invaluable for addressing challenges such as climate change, resource limitations, and the need for sustainable agriculture.

Keywords Climate resilience · Food security · Nutritional security · Omics approaches · Sorghum

Introduction

There might be a food and nutritional crisis by 2050 due to the increasing population, and traditional crop management strategies would not be sufficient to address it. Thus, only the finest quality of these climate-resistant crops will be used to supplement the diet during that time (Jones et al. 1970). Scientists have gained a deeper grasp of the molecular mechanisms of plants by applying interdisciplinary methodologies. This has resulted in the development of more resilient and efficient crop varieties, which have the potential to significantly impact both sustainable agriculture and food security worldwide(Okoh et al. 1982). Food systems may be impacted by climate change in a number of ways, from direct effects on crop production (e.g., variations in rainfall causing drought or flooding, or

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variations in temperature causing the duration of the growing season), to changes in markets, food pricing, and food availability. One of the most sensitive indicators of climate change is thought to be plant phenology, which is the cyclical growth and development of vegetation in the natural environment. Phenological changes have a significant effect on the water and carbon cycles (Xue et al. 2023). Studying environmental changes and their effects on ecosystems can yield important information for developing sustainable agricultural methods, such as those pertaining to the production of sorghum (Wang et al. 2022). For instance, one of the most often mentioned factors influencing food insecurity in southern Africa is climate insecurity because it functions as a transient shock as well as an underlying, persistent problem (Gregory et al. 2005). A move towards less established and underutilized, climate-resilient crops like sorghum is very interesting in addressing this issue. When it comes to temperature and water scarcity, sorghum performed better than the three main cereal crops viz., wheat, rice, and maize and corn (Teferra 2019). In an effort to draw attention to wholesome grains, the UNGA (United Nations of General Assembly) proclaimed 2023 to be the International Year of Millets (IYM 2023) in March 2021. According to the most recent data from the Food and Agriculture Organisation of the United Nations (FAO), the world's production of sorghum reached over 61 million tonnes in 2021, encompassing roughly 41 million hectares of arable land (Lee et al. 2023). With a production output of more than 29.7 million tonnes, Africa led the globe in sorghum production in 2017. According to FAOSTAT's data, after maize, sorghum is the most widely grown cereal crop in Africa and ranks sixth in the world in terms of production volume in Sub-Saharan Africa. Because it uses the C4 carbon fixation pathway to boost the efficiency of its photosynthetic activity, sorghum is recognized as a C4 crop(Adebiyi et al. 2005). It flourishes in hot, semi-arid, tropical climates that are prone to drought and have lesser rainfall.

Sorghum (*Sorghum bicolor*), is a well-known crop that resists drought and changes in the climate and provides a significant portion of the food consumed in Africa and other developing nations. Compared to other major cereal crops, sorghum has a higher proportion of slowly digested and resistant starch and many other nutritional components, which is a major advantage that makes it a healthy and nutritious crop (Table 1). This characteristic of sorghum lowers postprandial hyperglycemia in people, and it may be used to control the total amount of calories consumed from sorghum-based goods. The inherent advantages of this crop and its use as a regular meal highlight the significance of this crop's transformation into an edible form. Not only do these meals have cholesterol-lowering, probiotic, antioxidant, and antibacterial properties, but they are also known to have beneficial benefits that are both useful and therapeutic(Cheatham et al. 2017). Sorghum is also a source of several significant bioactive chemicals (Udachan el al. 2012). As a result, meals made from fermented sorghum possess a lengthy history and a close cultural affinity with Africans. Sorghum is a member of the *Andropogoneae* tribe and *Poaceae* family and is a well-known C4 crop (i.e., it increases photosynthetic efficiency using the C4 carbon fixation pathway), which is specially adapted to hot, drought-prone, and semi-arid tropical conditions with little rainfall (Oluwafemi 2020). An effective method for creating superior cultivars is to use omics technologies, such as genomics, proteomics, transcriptomics, and metabolomics (Fig. 1). There are now more options to better understand and view the quality of fermented foods from a variety of

Table 1 Quantitative analysis of biochemical constituents present in Sorghum

Sr. no.	Ash	CHO	Fat	Fiber	Moisture	Protein	References
1	1.2–1.3	NR	3.1–3.4	1.8–1.9	NR	11.5–11.7	Jones and Beckwith (1970)
2	0.90–1.52	71.80–85.20	1.38–4.50	1.47–2.45	NR	9.28–14.86	Okoh et al. (1982)
3	1.98	72.41	3.35	2.25	10.66	9.35	Adebiyi et al. (2005)
4	4.20	NR	6.9	19.5	8.1–8.5	11.80	Shawrang et al. (2011)
5	1.44	NR	3.32	1.83	NR	9.95	Kumar et al. (2022)
6	0.92–1.75	70.65–76.20	2.30–2.80	1.40–2.70	8.10–9.99	8.90–11.02	Udachan et al. (2012)
7	1.28–1.78	72.44–77.28	2.84–3.02	1.72–2.02	6.67–7.29	10.21–13.45	Cheatham and Sheppard (2017)
8	1.61–2.03	NR	2.37–2.75	NR	8.95–11.16	11.90–12.82	Ndimba et al. (2017)
9	1.90	NR	3.30	1.7	9.80	12.5	Sharma et al. (2015)
10	2.07	76.51	3.10	2.86	6.36	9.10	Ape et al. (2016)
11	1.12–1.68	65.15–76.28	5.12–10.54	1.65–7.94	1.39–19.02	6.23–13.81	Jimoh and Abdullahi (2017)
12	3.17	71.95	4.70	2.76	6.07	11.36	Oluwafemi (2020)

In dry matter, the values are expressed

CHO carbohydrate, *NR* not reported

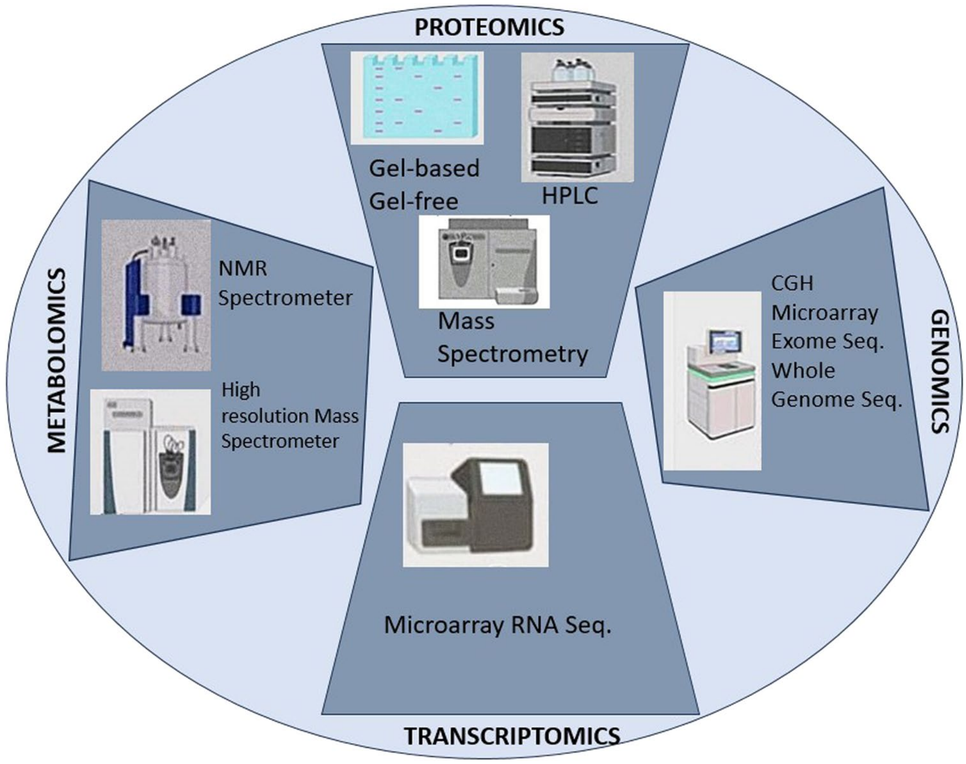

Fig. 1 Various techniques involved in genomics, proteomics, transcriptomics and metabolomics analyse

angles thanks to the advent of "omics" technologies, including genomics, metabolomics, metagenomics, foodomics, transcriptomics, volatilomics, and proteomics.

The identification and tracking of plant physiological reactions as well as stress caused by biotic and abiotic metabolic pathways or linkages can be done using metabolomics approaches (Ndimba et al. 2017). The different instruments used in metabolomics are high-performance liquid chromatography (HPLC), electrospray ionization-mass spectrometry (ESI–MS), gas chromatography-mass spectrometry (GC–MS), and gas chromatography (Sharma et al. 2015). Transcriptome study reveals that the response of maize to drought stress involves alternative splicing, transcription regulation, and hormone metabolism(Ape et al. 2016). It can be discovered through various applications BSTA, qRT-PCR validation, and QTL association investigations. Numerous scientists are now focusing on the topic of proteomics to examine how different stressors affect physiological parameters differently at the proteome level (Kumar et al. 2022). Different molecular biology techniques, including RNA sequencing, microarrays, and serial analysis of gene expression, are used to quantify the study of differential expression. While the microarray determines the number of designated transcripts, SAGE, and qRT-PCR technologies, RNA sequencing makes use of the benefit of high-throughput sequencing to identify novel transcripts. Specific metabolites that are present in a sample are identified and quantified using metabolomics(Zhang et al. 2010). Quantification of physiologically active chemicals can be aided by metabolomics, food profiling, and food fingerprinting nuclear magnetic resonance (NMR), liquid chromatography coupled to mass spectrometry (LC–MS), inductively coupled plasma (ICP), and gas chromatography coupled to mass spectrometry (GC–MS) (Fig. 2). A nutritionally dense, diversified, and balanced diet for society is the goal of current crop research methods using integrated omics platforms. The researchers have been able to increase the amounts of important nutrients in agricultural plants by knowing and regulating many scientific research areas, such as genomics, proteomics, and metabolomics (Nayak et al. 2021). The objective of this review article is to overview sorghum's nutritional value and possible utilization of multi-omics approaches in the growth, development, and stress resilience of this nutritionally rich climatic resilient plant.

Sorghum genomics

Genomics is a biological field that studies the structure, function, evolution, mapping, and editing of genomes. A genome is an organism's complete DNA, including all of its genes and its structural configuration. Functional studies, which yield information that may be the most immediately useful for crop development, are often used to realize genomics research. To understand gene functions and the relationships between genes in regulatory networks,

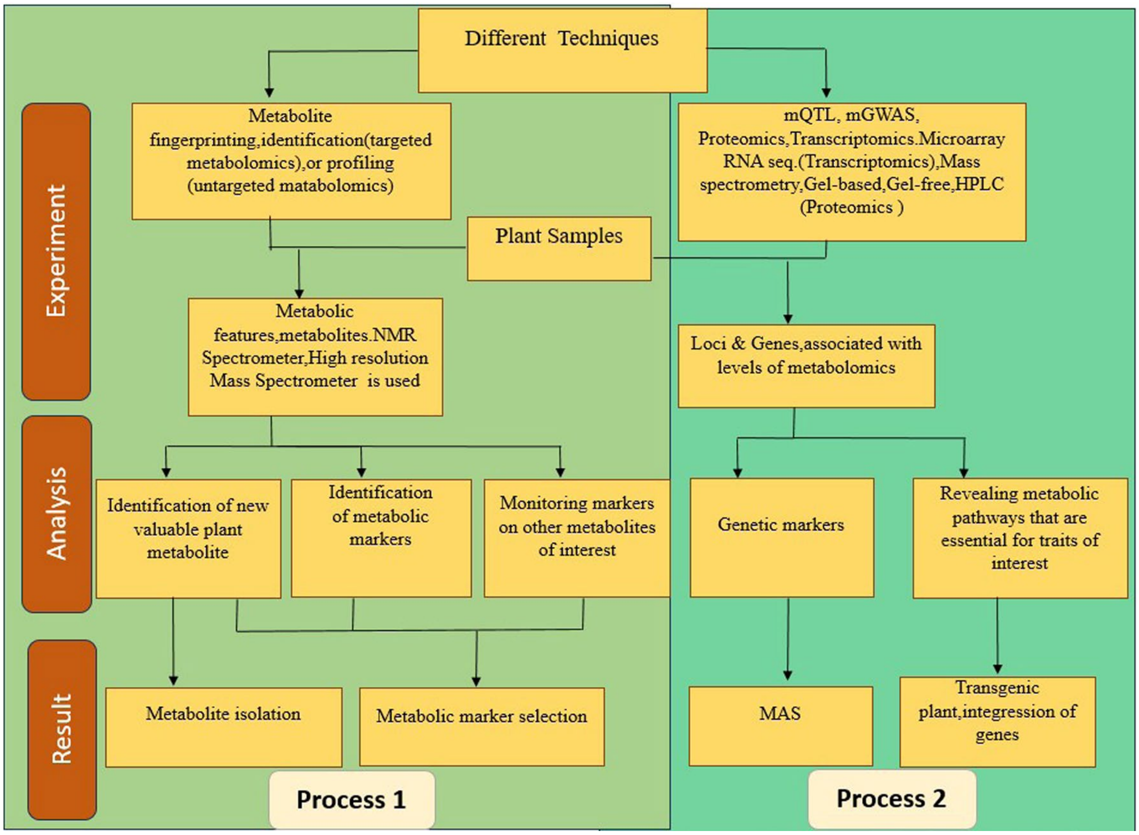

Fig. 2 General mapping and integration can be done with QTL mapping and GWAS. Plant samples (roots, leaves, grains, etc.) are collected with care for metabolic analysis. Plant samples are typically taken at several growth stages, from various crop accessions, or even from the same accessions cultivated under various environments. In Process 1, metabolomics research is applied. The result of such studies gives the metabolites or the metabolomics feature. Process 2 assumes the integration of genetic information. The metabolite concentrations obtained in the first step can be used as quantitative traits for QTL mapping analysis (which requires a defined breeding protocol for generating plant samples) or GWAS (which requires a substantial number of genetically diverse plant samples of the cultivar of interest). It results in the identification of metabolic QTLs. Metabolic QTLs may be used for the creation of genetic markers and their use in MAS. Additionally, in order to enhance crops, beneficial genes, and gene sets can be found in both related as well as unrelated plant species

which may be used to create better varieties, functional genomics techniques have long been used (Akpinar et al. 2013). There are some functional and structural genomics approaches which include techniques like Sequencing-Based Approaches, Hybridization-Based Approaches, Genome Sequencing, and Mapping. Some useful markers like SNPs (Single Nucleotide Polymorphism) are particularly useful in genomics because of their high abundance in genomes. When reference genome sequences or large transcript databases are available for a species, SNPs can be easily found by genome or transcript resequencing as well as genotype comparison. NGS (Next Generation Sequencing) is another helpful method. One of its primary benefits has been the significant contributions in NGS-mediated shotgun sequencing which leads to the development of robust molecular markers. These markers are more environment-independent than morphological markers, which were the primary focus of traditional breeding research, and they indicate diagnostic DNA sequence variations. These days, high throughput sequencing (HTP) inside sequence assembly techniques and measuring platforms are improving quickly and getting cheaper, making genomic tools and attributes for many crops easily accessible (Jayakodi et al. 2021). For the functional and comparative studies of the genome along with crop enhancement at the molecular level, various tools which include genetic mapping, DNA markers, and high-quality reference genomes were used (Zhang and Hao 2020). First-generation sequencing, or the traditional Sanger sequencing method, was improved by genome sequencing techniques (Li et al. 2017; Cui et al. 2020). With the help of next-generation sequencing and third-generation sequencing, numerous crops were sequenced (Zenda et al. 2021). Sorghum (*Sorghum bicolor*), is renowned for both its exceptional biological qualities and agronomic traits, including a high tolerance for heat and drought stresses. It is spread over 8 million hectares across the global farmland which contributes to fiber, feed, food, and fuel. Because of this, it is diversified genetically in various levels as it is divided into different landraces, of which sweet, forage, and grain sorghums are commercially grown worldwide. It is a promising source for producing bio-fuel that is used for bio-ethanol production as well as bio-energy crop due to very high sugars, starch, cellulose, lignin, hemicellulose, and grease content, with its ability of stress tolerance, fast growth, and good yield (Tian et al. 2016).

Sorghum has a small, non-duplicating genome, and is diploid in nature. Sanger sequencing was first used for the sequencing of the whole genome of *Sorghum bicolor* BTx623 which is also a reference genome (Paterson et al. 2009) and modified by McCormick et al. (2018). The modification consists of approximately 34,129 annotated genes, 8.7% of DNA transposons, and about 58.8% of retrotransposons. As the sequencing technologies evolved, many other genome sequences of sorghum have been conveyed. However, to know the phenotypical alteration's fundamental genetic processes or mechanism while growing sorghum and enhancement of this crop, there is a need for more in-depth research, which can bring more opportunities in molecular breeding and better genomics research in sorghum (Hao et al. 2021). Research has led to the development of a comprehensive platform known as the "sorghum database," or sorghum genomic functional database (Sorghum FDB), which can assist researchers in finding the complete functional annotation of sorghum. This platform is made up of data from public resources and data found in related databases (http://structuralbiology.cau.edu.cn/sorghum/) (Tian et al. 2016).

Genes linked to drought tolerance

One major environmental factor limiting plant productivity is drought. Being a complex stress, it causes a broad range of physiological, biochemical, and molecular reactions in plants (Kaur and Asthir 2017). From an agricultural perspective, a drought is characterized by below-average precipitation, fewer rainy days, or above-normal evaporation, which frequently leads to a decline in crop growth and output (Li et al. 2024). Drought stress causes stomatal closure, lowers transpiration and photosynthetic rates, leads crops to mature early, and results in low productivity (Kaur and Asthir 2017).

To handle the environmental stress conditions, plants have the tendency to change the gene expression which is related to various factors such as molecular, biochemical, and physiological. Genes can act under drought conditions and undergo upregulation and downregulation according to their mechanism when stress conditions are induced. In sorghum, *SbSNAC1* gene is stimulated during drought conditions (Lu et al. 2013), but most of the time, one solo gene is not enough to cope with the resistance to stress conditions (Roy et al. 2011). Hence, transcription factors act as key regulators for the expression of various genes which have dimensional functions in a plant's adaptation to stress conditions. Severe drought stress causes plants to need extra time to alter their internal osmotic potential to match their surroundings (Abreha et al. 2022).

Plants are the primary home of LIM domain-containing proteins, which are important for a number of biological functions including actin cytoskeletal structure and gene transcription. It has been observed that sorghum's *SbLIM1*, *SbLIM2*, and *SbLIM4* genes are closely related to *Zea mays.* The LIM gene family in sorghum will provide important insights about their roles in growth, development as well as defense against various biotic and abiotic stresses (Sarkar et al. 2023). The *AtHRD* contains domains like *AP2* (*APETELA2*) / *ERF* (ethylene responsive factor) is related to drought resistance, as *AtHRD* attaches to the GCC box of genes related to drought and acts in drought tolerance (Shanker et al. 2012). Other than these, there are transcription factors involved in abiotic stress which consist of bZIP, zinc fingers, MYB, bHLH, and GRAS get upregulated in sorghum in drought conditions (Abdel-Ghany et al. 2020). Other transcription factors i.e., Dof, which stands for DNA binding one finger, are recognized to have a regulatory function in drought resistance of sorghum. When drought conditions occurs, some of the *SbDof* genes are upregulated and some get downregulated. The genes that are upregulated are *SbDof12*, *SbDof*19, and *SbDof*24 whereas, the genes that are downregulated are *SbDof6*, *SbDof*15, *SbDof*16, *SbDof*18, and *SbDof*20. *SbDof* genes also show their expression on different phases of drought conditions as *SbDof19* and *SbDof*24 are expressed at maximum level at the beginning of the drought whereas, *SbDof21*, *SbDof*22, *SbDof*23, *SbDof*25, *SbDof*27, and *SbDof*28 got upregulated at post-drought stages (Gupta et al. 2016). *SbDof* genes play a significant supportive function in drought tolerance in sorghum plants as these genes also showed time-dependent variation in expression. Auxin-responsive genes are regulated by *ARF* (auxin response factor), which includes *GRETCHEN HAGEN3* (*GH3*) and *LATERAL ORGAN BOUNDARIES* (*LBD*). Expression of these genes gets upregulated highly when exposed to drought-like conditions which shows the function of auxin in drought response (Wang et al. 2010; Gupta et al. 2016). NAC (*N*-acetyl cysteine) transcription factors get regulated under abiotic stress conditions while activating the transcription of various other genes. In drought stress conditions of sorghum, *SbNAC* genes like *SbNAC014*, *SbNAC034*, *SbNAC035*, *SbNAC037*, and *SbNAC041* showed enhanced expression, whereas, *SbNAC116*, *SbNAC073*, and *SbNAC052* got downregulated at the similar time (Sanjari et al. 2019). Transcription factor, or the *WRKY* gene family, has been demonstrated to play a critical part in the plant's ability to adapt to stressful situations, such as drought. A

Table 2 Gene expression profiling of *Sorghum bicolor* during drought stress conditions

Sr. no.	Genes	Upregulated/ downregulated	Function	References
1	*SbNAC1*	Upregulated	Acts in drought tolerance	Lu et al. (2013)
2	*SbNAC052, SbNAC073, SbNAC116*	Downregulated	Negative regulation of expression of stress-responsive genes	Sanjari et al. (2019)
3	*SbNAC034, SbNAC035, SbNAC041*	Upregulated	Transcriptional activator, functions in post-flowering drought stress conditions	Sanjari et al. (2019)
4	*SbNAC037*	Upregulated	Remobilization of nutrients during drought stress	Sanjari et al. (2019)
5	*SbDof12, SbDof19, SbDof24*	Upregulated	Helps in the tolerance during the drought stress condition	Gupta et al. (2016)
6	*SbDof21, SbDof22, SbDof23, SbDof25, SbDof27, SbDof28*	Upregulated	Functions as tolerance factor at a later stage of the drought stress condition	Gupta et al. (2016)
7	*SbDof6, SbDof15, SbDof16, SbDof18, SbDof20*	Downregulated	Negatively expressed at the time of drought stress conditions	Gupta et al. (2016)
8	*SbGH3 SbLBD*	Upregulated	Drought stress signalling	Zhang et al. (2010); Gupta et al. (2016)
9	*SbWRKY30*	Upregulated	Highly expressed in the taproots and leaves of sorghum	Yang et al. (2020)
10	*SbWRKY45, SbWRKY79, SbWRKY83, SbWRKY16*	Upregulated	Responsible in the drought stress response during seedling, flowering, and dough stages	Baillo et al. (2020)
11	*AtHRD*	Upregulated	Attaches to the GCC box of genes related to drought and functions in drought resistance	Shanker et al. (2012)
12	*Bzip, zinc fingers, MYB, BHLH, GRAS*	Upregulated	Drought tolerant genotype	Abdel-Ghany et al. (2020)

novel *SbWRKY30* gene, which gets expressed at regions like taproot and the leaf part of sorghum in drought conditions and particularly joins to the element called W-box which is present in the promoter region of *SbRD19* gene which is a drought stress-responsive gene of sorghum (Yang et al. 2020). In addition to the *SbWRKY30* gene, other members of the *SbWRKY* family, such as *SbWRKY45*, *SbWRKY*79, *SbWRKY*83 and *SbWRKY*16, are also highly expressed in sorghum under drought stress (Table 2). Because of this, the total expression of *SbWRKY* genes under drought conditions is crucial for drought tolerance (Baillo et al. 2020; Rajendra Prasad et al. 2021).

Gene expression during high-temperature stress

Many researchers are interested in plants that can withstand heat stress than ever before, mainly because of worries about how climate change may impact natural and managed ecosystems. Radiative forcing is the word used to describe the increase in energy in the atmosphere caused by rising greenhouse gas concentrations (Jagadish et al. 2021). Initially, while the term "heat stress" is used widely, it includes warming studies, heat shock, and heat waves, all of which have different durations and intensities of induced temperature increases. To get a better understanding of how plants will be affected by the heat stress associated with global warming, there is a need of stronger integration of the outcomes and tools across different approaches.

The transcription factors like bHLH (basic-helix-loop-helix), PIF4 (phytochrome interacting factor 4) were induced when high-temperature stress was observed in sorghum and some other orthologs of these transcription factors were also found in various other crops along with sorghum. The function of these genes during high-temperature stress conditions was demonstrated by Proveniers and Van Zanten (2013). Other than these genes, auxin-responsive factors i.e., *ARF* genes also expressed in sorghum during high-temperature stress. When sorghum crop gets induced with high temperature, the expression of 9 *SbARF* genes got upregulated. The genes that are active when high-temperature stress is induced are *SbARF1, 5, 6, 8, 10, 11, 12, 16, 18, 20,* and *25*. The genes that are downregulated under heat stress are *SbARF4, 7, 9, 15, 17, 19, 21, 22,* and *24* (Chen et al. 2019; Rajendra Prasad et al. 2021) (Table 3).

Genes regulate both drought and high-temperature stress

During drought and high-temperature stress, some signaling pathways get activated and stimulate the expression of some genes that help the plant endure under worst/ unfavorable conditions. Signaling chemicals such as nitric

Table 3 Differential expression of genes in *Sorghum bicolor* during high-temperature stress conditions

Sr. no.	Gene(s)	Upregulated/downregulated	Function	References
1	*bHLH, PIF4*	Upregulated	Gets activated during heat-stress conditions	Proveniers and Van Zanten (2013)
2	*SbARF4, SbARF7, SbARF9, SbARF15 SbARF17, SbARF19, SbARF21, SbARF22, SbARF24*	Upregulated	Functions after the high-temperature stress conditions	Chen et al. (2019)
3	*SbARF1, SbARF5, SbARF6, SbARF8, SbARF10, SbARF11, SbARF12, SbARF16, SbARF18, SbARF20, SbARF21*	Upregulated	Acts during the high-temperature stress conditions	Chen et al. (2019)

oxide (NO), sugars, CDPKs, MAPK/MPKs, and specific plant hormones are important signaling pathways that trigger the expression of many genes that react to stressful conditions (Ahmad et al. 2012). Mitogen-activated protein kinase i.e., MNPK pathway triggers DSM1 (drought-hypersensitive mutant 1), acts for the tolerance to drought stress when the plant is under stress conditions. Reactive oxygen species (ROS) are cleaned up or detoxified by stress-responsive genes that are triggered. This preserves cellular homeostasis, activates structural proteins and enzymes, and eventually leads to tolerance to abiotic stresses, which include drought and high temperatures (Ciarmiello et al. 2011). HSFs i.e., heat shock transcription factors, control the expression of heat shock proteins (HSPs) which act as an important factor when stress is induced. Various HSPs are expressed when high temperature or drought stress is induced in sorghum. For instance, *HSF1* was highly expressed during high-temperature stress, while some got upregulated in drought stress conditions like *HSF5*, *HSF6*, *HSF10*, *HSF13*, *HSF19*, *HSF23*, and *HSF25* (Nagaraju et al. 2015). In some genotypes of sorghum, heat shock factors like *SbHSF5* and *SbHSF13* got upregulated in leaf tissues under drought-stress conditions. Glutathione reductases (GRs) are also recognized to act as essential role in antioxidant machinery. Rho-type GTPase-activating protein (RGA1) also plays an important signaling molecule function in abiotic stress, with other cis-regulatory elements, which are ABA, ARE, GT-1 boxes, LTR, and MeJAE. RGA1 has shown its upregulation under drought stress conditions while downregulation under high-temperature stress conditions (Rajendra Prasad et al. 2021). Genomics plays a crucial role in the development and improvement of sorghum plants. Genomics approaches advancing sorghum plant development by providing insights into the genetic basis of traits, facilitating breeding strategies, and accelerating the development of improved varieties with enhanced agronomic and economic characteristics.

Sorghum proteomics

Proteomics is the field that involves extensive analysis of the proteome of an organism, cell, or tissue, while the growth and development of an organism or the changes that occur due to variable environmental conditions. The study of the entire proteome aspires to the idea of protein diversity, the amount of protein present, its isoforms, location as well as its interactions with other protein molecules and post-translational modifications (PTMs) (Labuschagne 2018; Kosova et al. 2018). It has been established by earlier research that there is little to no direct association between the transcriptional stage of mRNA expression and the phenotypic of the plant. On the other hand, proteins act as effector molecules in plants that respond to developmental and environmental changes, and because of this, proteomics is stated as an important connection between transcriptomics i.e., the study of mRNA transcript, and metabolomics i.e., the study of metabolic pathways (Tan et al. 2017; Labuschagne 2018). Furthermore, the proteome is dynamically considered to the genome which is inert, and modifications in the expression of proteins give rise to post-transcriptional modifications, which thus provide more insight to know the function of protein biologically (Wu and Wang 2016).

In the past few years, there has been a significant increase in the analysis of plant proteomes. Protein identification is now comparatively simple due to advancements in bioinformatics, mass spectrometry technology, stains, and softwares (Quirino et al. 2010). The main methodology for comparative quantitative proteomic studies in numerous plant species has been the application of traditional two-dimensional electrophoresis (2-DE) techniques (Neilson et al. 2010). Without labeling restrictions, researchers will be able to analyze changes in protein expression patterns in different cell types due to the combination of spectral counting and shotgun proteomics. Therefore, it is acceptable to apply such methods for analyzing numerous data points in a quantitative investigation of abiotic stress in plants, including a range of temperature readings or periods of time. Numerous strategies for performing proteomics studies have been reported in many cereal crops. These strategies include mass spectrometry, data processing techniques, and gel-based or gel-free techniques for protein identification coupled with mass spectrometry (Fig. 3). The gel-based approach of proteomics consists of 1/2 dimensional PAGE i.e., 1 or 2-D polyacrylamide gel electrophoresis, and the differential in gel electrophoresis (DIGE) techniques for protein separation, along with quantification of the protein, its digestion, and identification through mass spectrometry (Tan et al. 2017; Labuschagne 2018). Contrary to gel-based techniques, gel-free-based techniques can also be used which consists of intact proteins digestion by the protease degradation into smaller molecules of proteins called peptides and separated using liquid chromatographic technique and identified using mass spectrometry, consisting of isobaric tags used for iTRAQ (relative and absolute quantitation) ICAT (isotope-coded affinity tags) and TMT (targeted mass tags) (Tan et al. 2017). Over the course of two decades, there has been a significant advancement in plant proteomics, which can be summed up as the improvement of conservative techniques and the beginning of novel, high throughput, and high-resolution approaches. These approaches include sample preparation, protein extraction, fractionation, quantification, and analysis procedures, as well as data processing and proteomic study analysis (Tan et al. 2017). The field of proteomics has continuously evolved from ordinary descriptive studies of numerous plant proteins and covalent changes to in-depth analyses of PPIs (protein–protein interactions) and PMIs (protein-metabolite interaction) (Ramalingam et al. 2015; Scossa et al. 2021). Because of the advancement and new evolutions in the system of liquid chromatographic-tandem mass spectrometry which have notably enhanced the scanning rates and also the resolution. Specifically, the area

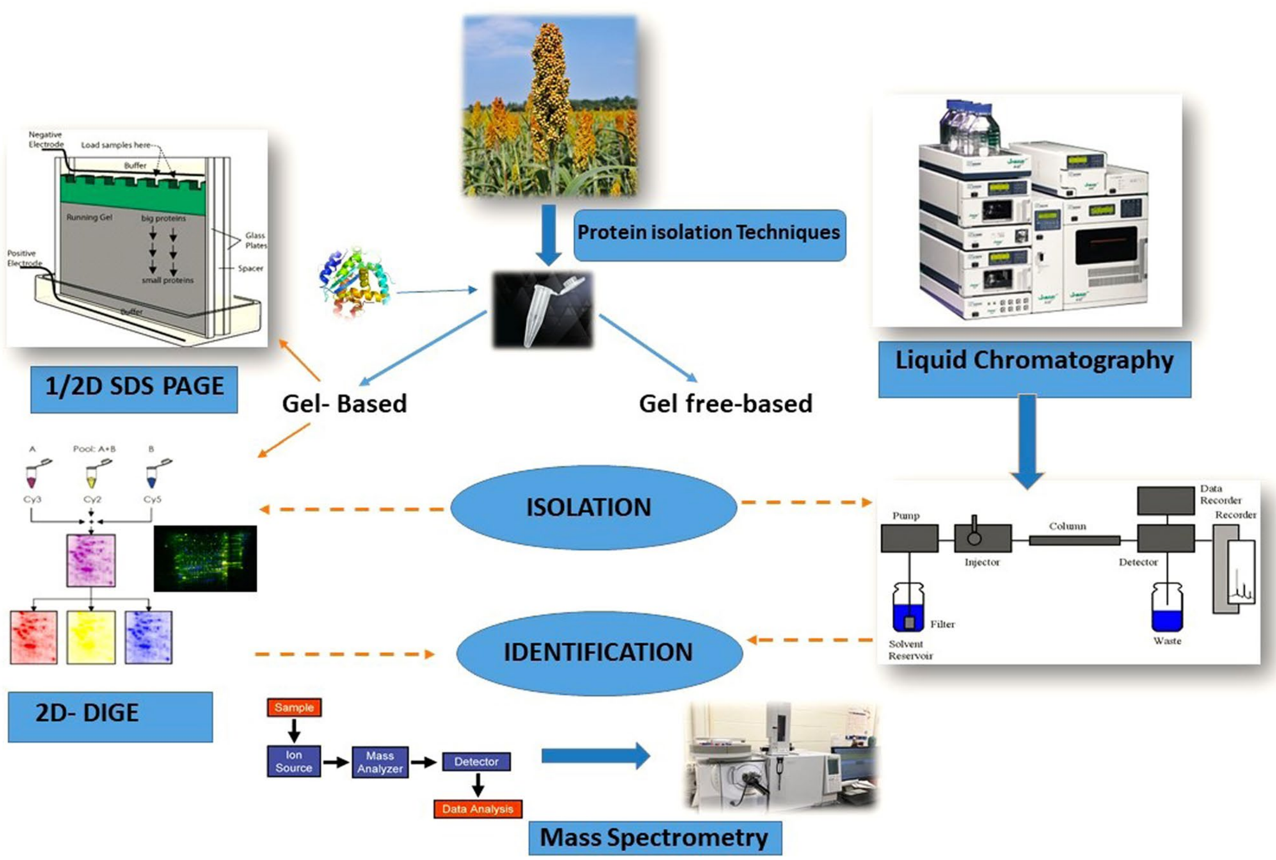

Fig. 3 Various proteomic techniques used for qualitative and quantitative analysis of Sorghum. Here, 1D SDS-PAGE is a straightforward technique for separating proteins based on size, while 2D SDS-PAGE adds an additional dimension of separation based on charge, providing a more comprehensive analysis of complex protein mixtures. 2D-DIGE is a powerful and quantitative technique that enhances the capabilities of traditional 2D gel electrophoresis by providing a reliable and reproducible method for comparing protein expression levels between different samples. Liquid chromatography (LC) is a widely used analytical technique that separates, identifies, and quantifies components in a mixture. It is based on the principle of selective retention of compounds in a liquid mobile phase as it moves through a stationary phase. Mass spectrometry is an analytical technique used to identify and quantify the chemical composition of a sample based on the mass-to-charge ratio of charged particles. It involves ionizing the sample, separating the ions based on their mass-to-charge ratio, and then detecting and measuring the abundance of these ions

of protein-metabolite interactions (PMIs) has been focused to understand the function of metabolites, rather than functioning as metabolic intermediates. They also act as ligands or co-factors, which has the ability to modify the conformation and function of the proteins (Scossa et al. 2021; Zenda et al. 2021).

Protein expression during drought stress

When stress gets induced, proteins that are encoded by the genes of sorghum, activate the signaling pathways, regulating the expression as well as the translocation of the proteins to any relevant stress. To know the plant's response to different stresses, and the proteins that act as key factors to deal with and bring out the resistance to various stresses that act on plants, the study of proteomics has shown its importance. Some of the recent studies conducted in the field of proteomics, give us better insight to know the molecular-level response of the plant to various stresses including drought stress (Ngara et al. 2021). In a proteomic analytic study, E19 a drought-tolerant genotype of sorghum, and a sensitive genotype of sorghum were exposed to drought-like conditions, which resulted in the expression of GrpE protein homolog, *HSPs*, and GR-RBP (glycine rich-RNA-binding protein) (Fadoul et al. 2018). GrpE, which is a stress-inducible protein regulates *HSP* and gets expressed under drought stress conditions in other crops, as well (Kim et al. 2015; Piveta et al. 2020). GR-RBP demonstrated its ability to regulate ABA and the stress response in addition to acting on RNA transcription (Kim et al. 2010). Another study found that, when drought stress was generated, certain *HPSs* and dehydrins were elevated in two genotypes of sorghum i.e., BTx642 (sensitive) and RTx430 (pre-flowering drought tolerant). The *HSP* 60 (chloroplastic form) and di-sulfide isomerase got upregulated in drought tolerant genotype after 24 h of recovering from the exposure to drought stress (Jedmowski et al. 2014). Organelles can be more genotype-specific than cytosol, hence the proteome study comparing the cytosolic and organelle from leaf sample of two sorghum genotypes, *Sorghum bicolor* RTx430 and *Sorghum bicolor* RTx642 showed responses to stress brought on by drought revealed distinct responses. In a study concluded by Ogden et al. (2020), proteins associated with ABA metabolism, signal transduction, Rubisco gene activation, ROS (reactive oxygen species) scavenging, flowering time regulation, and epicuticular wax synthesis were found to be localized in organelles and are elevated in genotypes resistant to drought. ABA is a well-known stress hormone produced first during drought stress (Roychoudhury et al. 2013; Sah et al. 2016). Root proteome analysis of two different sorghum genotypes, drought-sensitive variety (ICSB338) and drought tolerant variety (SA1441), has revealed the proteins that react to water stress and has also revealed increased expressions of key ROS metabolism-related proteins such as glutathione-S-transferases, peroxidases, thioredoxins, and germins. Some transcription-related proteins including 6 histone proteins, and 3 NAC (nascent polypeptide-associated complex) subunit beta proteins, only got expressed in the root of drought tolerant genotype of sorghum at time of water scarcity (Goche et al. 2020). One of the most important transcription factors that contributes to plant senescence and aids in the tolerance of drought stress in plants is NAC (Wu et al. 2016). At post-flowering drought stress, thirteen proteins act to bring tolerance to the stress as plants respond to stress, which is expressed by the NAC transcription factor gene family (Sanjari et al. 2019). The analysis of proteome in response to drought stress in the root of sorghum in sapling stage has shown that proteins are related to the change in use of energy, osmotic alteration, ROS scavenging, synthesis of protein, processing of proteins as well as its breakdown by enzymes or other factors, has important function in the upholding of root growth during drought stress conditions (Li et al. 2020). Under post-flowering drought stress, leaf tissues showed upregulation of some defense and immunity-related proteins e.g. RuBisCo (ribulose bisphosphate carboxylase), acidic endochitinase and oxygen-evolving enhancer protein, in sorghum genotype BTx642 (Woldesemayat et al. 2018; Abreha et al. 2022). A proteomic study profiling revealed differential proteins responsive to salt stress in sorghum leaves identified upregulated proteins, such as glutathione S-transferase (XP 002458541, XP 002465442), peroxidase (XP 002463451, XP 002463451, XP 002463451), and the universal stress protein (XP 002443333), which are involved in the detoxification of reactive electrophilic compounds (Tu et al. 2023).

Many sensitive proteins, some of which are involved in stress and defense, photosynthesis, the production of small molecules, the metabolism of amino acids, and translational and catalytic regulatory activities, were activated by pest invasion. More specifically, proteins related to defense mechanisms, such as pathogenesis-related protein 5 (PR-5), chitin-binding-type-1 domain-containing protein, calmodulin, osmotin, thaumatin-like pathogenesis-related protein 1, peroxidases, glutathione *S*-transferase, non-specific lipid transfer protein, expansin-like EG45 domain-containing protein, abscisic acid stress ripening 3, trypsin inhibitor/alpha-amylase inhibitor are the recognized proteins that showed resistance against the pest and pathogens and strengthening the defense mechanism of plants (Tamhane et al. 2021). Proteomics provides a comprehensive understanding of the protein composition and dynamics in sorghum plants. This knowledge is essential for advancing our understanding of sorghum biology, improving stress resilience, and developing varieties with enhanced nutritional quality and agronomic traits.

Sorghum metabolomics

In comparison to other omics, metabolomics is a more recent field, and it is just now starting to contribute to crop development. Metabolomics must discover the biochemical networks underlying the targeted features in addition to metabolic indicators. Metabolomics and biochemistry are comparable in this regard, although metabolomics employs a more thorough methodology. To evaluate the significance of biochemical knowledge to crop improvement, commercialized genetically modified (GM) crops with modified metabolic activities were selected from the GM Approval Database, which is kept up to date by the International Service for the Acquisition of Agri-biotech Applications (ISAAA). In contrast to genetic approaches, metabolomics techniques require a great deal of care in sample preparation and expensive equipment, even for relatively simple tasks. The fact that they are utilized significantly less frequently than genetic techniques is not surprising. With thousands of articles published each year, metabolic investigations are, nevertheless, increasingly becoming more widespread. Purely metabolic research and studies utilizing alternative ohmic methodologies are the two basic categories into which these investigations may be split. Therefore, metabolites can serve as helpful indicators of crop characteristics that are essential for agronomy. Moreover, QTL (Quantitative Trait Loci) and GWAS (Genome-Wide Association Studies)have a significant influence on the progress of functional genomics. It is projected that the process of using metabolomics technologies in crop breeding will be enhanced (Litvinov et al. 2021).

In the developing field of "omics" called metabolomics, metabolites and the chemical imprints process of cellular regulation in various biological species are identified and quantified. A crop used for manufacturing ethanol and a source of bioenergy is sweet sorghum. The price of ethanol production from sweet sorghum is 20–56% less expensive than from sugar cane. The study of environment-gene interactions, characterization of mutants, phenotyping, the finding of biomarkers, and the development of new drugs are all significantly aided by metabolomics. Advanced metabolomics analytical techniques, such as high-performance liquid chromatography (HPLC), gas chromatography-mass spectrometry (GC–MS), non-destructive nuclear magnetic resonance spectroscopy (NMR), and to accelerate metabolic profiling, direct flow injection (DFI) mass spectrometry were used. To investigate disease resistance, robust ecotypes, abiotic stress tolerance, and metabolically aided crop breeding, metabolomics is an important tool. The production of a wide variety of metabolites has a significant impact on plant growth and development under various environmental conditions (Razzaq et al. 2019). Primary (central) metabolites and secondary (specialized) metabolites are the two categories into which all metabolites fall (Litvinov et al. 2021). Environmental metabolomics is the study of how plants interact with their surroundings. In the kingdom of plants, where there is a wide variety of metabolites, nearly 2,00,000 compounds have been identified, the bulk of which are unknown. It has been estimated that in various plant species, 10,000 secondary metabolites have been discovered. Each and every metabolite discovered shows structural differences with respect to the biochemical properties as well as the functions of the metabolites and those are assumed to be most important in the plant sciences (Razzaq et al. 2019). By identifying metabolites in response to environmental influences, it seeks to understand the potential effects of abiotic/biotic stresses on any crucial biochemical process. The plant metabolome also includes several specific secondary metabolites that provide tolerance to biotic/abiotic stressors, such as phenolics (10,000), alkaloids (21,000), and terpenoids (> 25,000) (Razzaq et al. 2019). Three different approaches, targeted metabolomics, semi-targeted metabolomics, and untargeted metabolomics can be referred to as metabolomics, metabolic, or metabolite profiling. The metabolome includes a collection of relatively minor molecular weight primary and secondary metabolites, as well as their precursors and metabolic pathway intermediates (typically 1500 Da) (Bueno and Lopes 2020). Primary metabolites are essential for growth and life-sustaining processes. Compared to primary metabolites, secondary metabolites have a greater impact on how plants react to stress, diseases, and other environmental stimuli.

Metabolomics in response to various stresses in sorghum

Nitrogen stress

Sorghum (*Sorghum bicolor* [L.] Moench), the world's fifth-most productive cereal crop, has high biomass yield qualities in some hybrids, making it a good candidate for long-term, cost-effective biofuel production. In nitrogen-stressed sorghum, special metabolic relationships with agronomic characteristics changed over the growing season. For crop development efforts to maximize the production of biofuel feedstock, a deeper comprehension of the metabolic responses to low nitrogen (N) will be beneficial. The biochemical differences between plants grown in the full N field and roots receiving low N was examined using a non-targeted metabolomics technique. Higher concentrations of asparagine, other amino acids, 2-imidazolidone-4-carboxylic acid (2-I-4-C), and allantoin were found in the roots from the entire N field. Higher biomass was also associated with roots that had lower levels of shikimic acid, a precursor in the production of aromatic amino acids; however, this correlation with biomass was very weak in the low N condition. Grain genotypes with high grain production have been found to be somewhat more susceptible to low nitrogen stress than genotypes with poor grain yield, even when subjected to the HN (High Nitrogen) treatment. While genotypes with high yields on HN are more susceptible to LN (low nitrogen) stress, this response is not always present, and certain genotypes are less impacted by LN stress (Grzybowski et al. 2022). It is observed that under the LN treatment, the length and width of the flag leaf both shrank by about 6.5 and 3.5%, respectively. Chlorophyll and nitrogen content showed greater differences, with LN treatment resulting in a decrease of 15.3% and 13.8%, respectively. Under the full nitrogen conditions, root metabolites associated with the Pseudomonas correlation reported the strongest negative association related to the plant defense hormone SA (Salicylic acid), finally revealing reduced defense response of the plant. Provided nitrogen stress to the roots of the plant carries the reduced phenylalanine, one of the major precursors for the SA production, and also provided some evidence of metabolic capacity to the defense responses under very low N conditions. The interaction between abiotic as well as biotic stresses might be affected the metabolic activity for the plant defense. It is essential for the breeding programs that become established for the biofuel production relation with the marginal soils (Sheflin et al. 2019).

Saline and alkaline stress

Increased interest in improving salt tolerance in crop plants has resulted from the fact that stress from salt and alkali reduces crop development and agricultural production worldwide. The basic processes underlying this tolerance are yet unknown. Sweet sorghum (*Sorghum bicolor* Linn. Moench), a monocotyledonous agricultural species, has a higher tolerance to salt-alkali stress than most other crops (Cui et al. 2022). Saline-alkaline soils reduce crop yield and production, which has a negative effect on agricultural productivity. The main sources of ions in alkaline soils are neutral and alkaline salts such as NaCl, Na_2SO_4, $NaHCO_3$, and Na_2CO_3. Particularly, the accumulation of Na^+ can cause a variety of osmotic and metabolic issues in plant shoots.

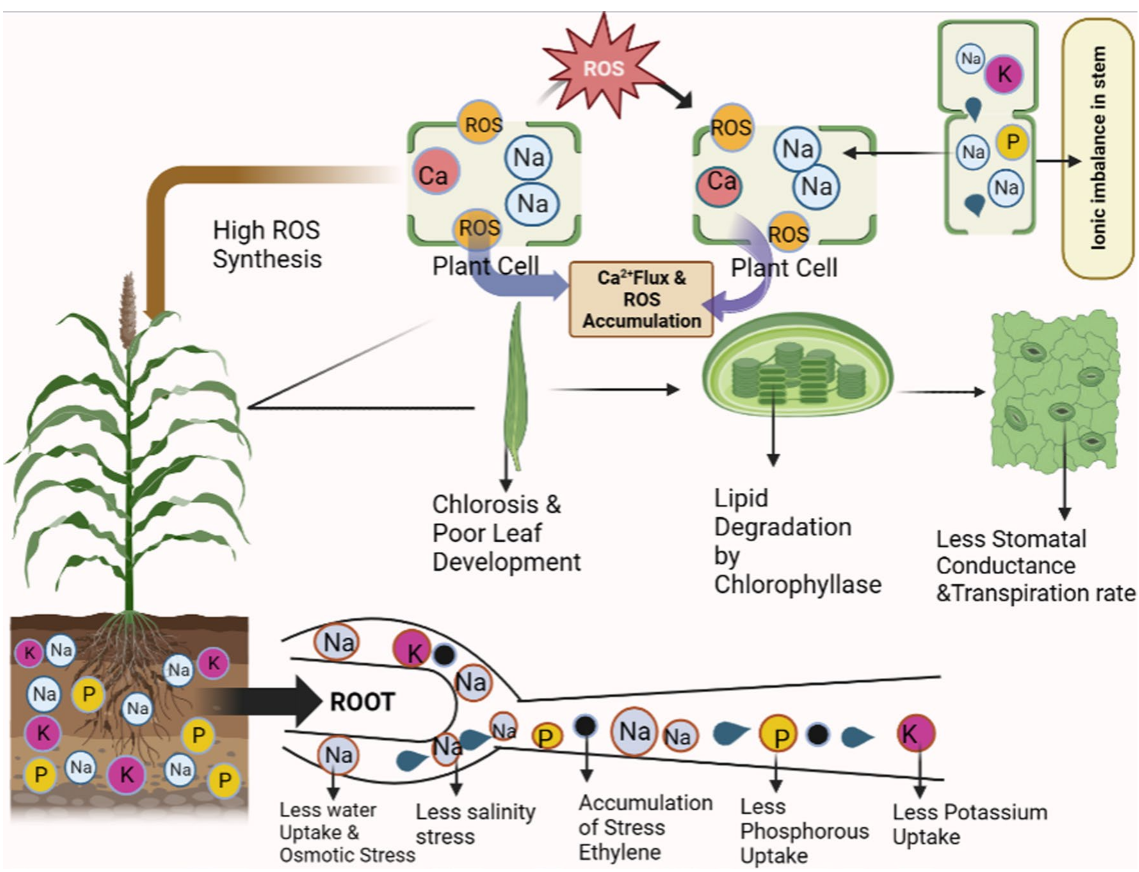

Fig. 4 Impact of ions stress and effect of ROS (reactive oxygen species) in Sorghum plant

It would appear that plant salt tolerance depends on inhibiting excessive Na^+ transfer to the shoots. One of the key variables in salt tolerance is the capability to sustain a high cytosolic K^+/Na^+ ratio, and salt storage in plant cell vacuoles is essential for this process. The free oxygen molecules in the cell would be targeted by the leaking electrons, which would cause an explosion of reactive oxygen species (ROS) and worsen the damage in the photosystem II (PSII) reaction center, even if it results in peroxidation or dissociation of the thylakoid membranes (Fig. 4). Sorghum is a C4 plant that is notable for its high yield, high photosynthetic activity capability, and robust resilience to environmental challenges such as drought, salt, and other conditions. The above-ground and underground parts of the plants responded differently to various forms of sodium salt. The fresh and dry weights of the underground region of sorghum were significantly lowered by the alkaline salts, especially Na_2CO_3. There was very little impact of the neutral salts on the subterranean portion of plant growth. Both the PSI (Photosystem I) and PSII response centers are frequently assaulted under stressful situations. According to several studies, PSI is more susceptible to damage from stress than PSII. The effects of several sodium salts, in the following order: $Na_2CO_3 > NaHCO_3 > Na_2SO_4 > NaCl$, were consistently seen in the development and photosynthetic function of sorghum leaves. When exposed to the higher alkalinity salt Na_2CO_3, the PSII reaction center's activity was more severely damaged. This suggests that the alkaline salt's relatively high pH value plays a role in the decreased PSII reaction center activity in sorghum leaves, in addition to the toxic effect of Na^+. Furthermore, sorghum leaf inactivated PSII reaction center ratio increased at high pH values (Zhang et al. 2018). Therefore, in addition to the effects of salt damage, soil alkalinity's impact should be considered when planting to encourage sorghum production in saline and alkaline terrain (Garcia-Caparros et al. 2021).

Defense-related changes through metabolomics

There are many diseases that affect the growth and development of sorghum but out of all diseases, one major disease namely bacterial leaf stripe disease is caused by the organism *Burkholderia andropogonis.* Numerous novel research methodologies have been created. When it comes to the features of molecular pathways related to pathophysiological interaction, metabolomics is a fundamental process. The previous findings showed a dynamic as well as intricate network of sorghum defense against *Burkholderia andropogonis* under the high defensive capability of the disease suppression (Mareya et al. 2019). Plants show a range of defense mechanisms against dangerous pathogens.

After analyzing the metabolic and molecular markers for the maize hybrid crop, it was determined that 130 metabolites were almost equal to the 38,000 SNPs (Single Nucleotide Polymorphism) in terms of predictive power (Riedelsheimer et al. 2012). Various studies were also conducted for metabolite inheritance pattern around various plant species like *Arabidopsis thaliana* (Rodriguez Cubillos et al. 2018), maize (Lisec et al. 2011), and foxtail millet (Zhang et al. 2018). The main aim was to determine the mode of inheritance, that it may be dominant, over dominant, additive, and/or non-additive. For the effective implementation of the biomarkers, there are two primary prerequisites, first one predictive ability/power to be robust through the different environmental conditions and the second one, developed biomarkers applicable in the various populations rather than the original population (Steinfath et al. 2010; Fernandez et al. 2016). Recent research provides extremely clear explanations of how metabolomics studies can be utilized to identify the signature biomarkers needed to distinguish between the plant cultivars that have recently been described (Chen et al. 2016; Song et al. 2018; Thomason et al. 2018; Zhang et al. 2018; Loskutov et al. 2020).

Metabolomics of seed color in sorghum

The taxonomy of metabolites consists of the primary and secondary metabolites that are essential for plant growth and development. The primary metabolites are responsible for carrying out life and growth processes but very few species have a high quantity of the primary metabolites, whereas, secondary metabolites are found in high quantities and involved in the many responses of the plant like biotic stresses as well as different forms of environmental stimulations. Among different classifications of metabolites, flavonoids play a key role in plants. Many different colored fruits, as well as flowers, possess very large amounts of flavonoids. These different types of flavonoids prevent plants against some damage throughout dormancy, UV light as well as phyto-pathogens. Fruits that are brilliant red and has anthocyanin and its derivatives, such as delphinidin, petunidin, malvidin, cyanidin, and pelargonidin, can take on purple and dark hues (Zhou et al. 2020). Plants of various colors have various forms of anthocyanins or flavones, which may be the chemical basis for their various colors. Metabolites are found in the seeds of sweet sorghum in an effort to identify the specific metabolites responsible for seed color development. Fruit color and flavor are determined by anthocyanins and flavonoids. Sorghum cultivars Z27 and HC4 have 217 significantly distinct metabolites, cvs. Z6 and HC4 have 240 metabolites, and Z6 and Z27 cultivars have 199 metabolites (Zhou et al. 2020). According to numerous prior research, secondary metabolites are mostly associated with changes in flower and seed colour having a big impact on how resistant an organism is invading stressors. Sixty-seven flavonoids had distinct expression patterns in Z6 and

HC4; the bulk of these expressions were in HC4, which has an expression pattern resembling that of Z6 and Z27. Anthocyanins are water-soluble plant pigments that are primarily responsible for the various hues (including white, red, black, and blue) seen in vegetables, seeds, fruits, and flowers. The metabolic diversity of three cultivars of sweet sorghum with white, red, and black seeds was examined and analyzed on different metabolites that would be connected with the seed colour and notably on the differential metabolites to better understand the processes that determine seed colour (Zhou et al. 2020). They observed a significant difference between the purple seed and the red seed color. Results depicted that, a total of 58 and 182 metabolites were downregulated and elevated, respectively, out of the 240 metabolites that were shown to be significantly different and associated with the purple colour. A total of 199 metabolites may be carrying the red phenotype related to 54 and 135 metabolites that were downregulated and upregulated, respectively. Their findings indicated that 45 metabolites exhibit all three cultivars and also *cyanidin 3-O-glucosyl-malonylglucoside*, *cyaniding O-acetyl hexoside*, and *cyanidin O-malonyl-malonylhexoside* were significantly up-regulated red seeds on the basis of a variety of the seed colors. Metabolomics is a powerful analytical approach that focuses on the comprehensive study of small molecules (metabolites) within a biological system. In sorghum, metabolomics provides valuable insights into the biochemical processes and pathways involved in growth, development, stress responses, and overall metabolism.

Sweet sorghum in response to salinity stress

It is believed that sweet sorghum has a high tolerance for saline. However, the existing genotypes of sweet sorghum are salt-tolerant as well as salt-sensitive. To maintain high levels of K^+ absorption, genotypes that are salt-tolerant are better able to reject harmful ions and hold the ingested hazardous ions in the root cell vacuoles. Therefore, it may be limited how much sodium and chlorine can build up in shoots and leaves that are actively growing. This system can successfully shield sweet sorghum's photosynthetic machinery from Na^+ and Cl damage. The high sugar content of sweet sorghum stems is maintained by a number of physiological processes. The mesophyll cells fix atmospheric CO_2 in a specific manner. The generation and accumulation of products created during photosynthesis are necessary for building up the sugar in the stalks of sweet sorghum. Oxaloacetate (OAA) is the first byproduct of the CO_2 fixation process. OAA is transformed into a form that can be transferred (malate), and it is then brought to the bundle sheath. 3-phosphoglycerate (3-PGA) is created through a sequence of events along with the C4 route, and it is later changed into triose phosphate (TP). Following production, TP either stays in the chloroplast stroma to complete the Calvin cycle or transform into starch, or it exits the chloroplast by exchanging orthophosphate for triose phosphate translocator (TPT). The cytoplasmic form of TP can be changed into fructose-6-phosphate (Fru-6-P). Sucrose phosphate phosphatase (SPS, EC3.1.2.24) or UDP-glucose (UDP-Glu) can then convert Fru-6-P to sucrose. Sucrose synthase (SS, EC2.4.1.13) is a reaction catalyst that uses UDP-Glu as one of its substrates. Invertase (INV, EC3.2.1.26) in the vacuole can break down sucrose into glucose and fructose. It is known that monocots include six sucrose transporters (SUT1-6), which are found in the tonoplasts of storage cells or the plasma membranes of sieve elements and companion cells. According to reports, SUTs are crucial to the re-distribution of sucrose (Sui et al. 2015).

Differentially expressed genes (DEGs) in response to salt stress

Salt tolerance in sorghum was regulated by a complicated mechanism by the identification of 4289 genes that were enriched in several GO (Gene Ontology) keywords and pathways (Liu et al. 2024). A comparison of sorghum cultivars M-81E and Roma without salt stress revealed that 3342 genes expressed at different levels. The DEGs between them, when salt was present, were 2265. 864 genes for M-81E had varied expression levels in normal and salt-exposed plants. 236 genes out of these DEGs showed increased expression in salt-stressed leaves. 930 genes for Roma had varied expression levels in normal and salt-treated plants. 442 genes out of these DEGs showed up-regulation in leaves during salt stress (Fig. 5).

Functional annotation using GO

To assign functional information to the DEGs separating NaCl-treated plants from control plants, Gene Ontology (GO) analysis was performed. The most significant DEGs or biological mechanisms that are overrepresented in reactions to drought stress were identified by GO enrichment analysis (Sui et al. 2015). In leaves, 190 common DEGs were affected by mild, severe, and re-watering treatments. 152 DEGs were enriched for 24 GO terms in the biological process and cellular component categories. This work provides a dynamic, regulated vocabulary together with hierarchical connections for the display of data on biological activities, molecular structure, and cellular components, enabling a uniform annotation of various gene products. In M-81E, 48 level-2 GO keywords were assigned to 812 different transcripts, which were grouped into three primary GO categories, thirteen for cellular elements, twelve for molecular operation, and twenty-three for biological processes. The 47 level-2 GO keywords in cultivar Roma

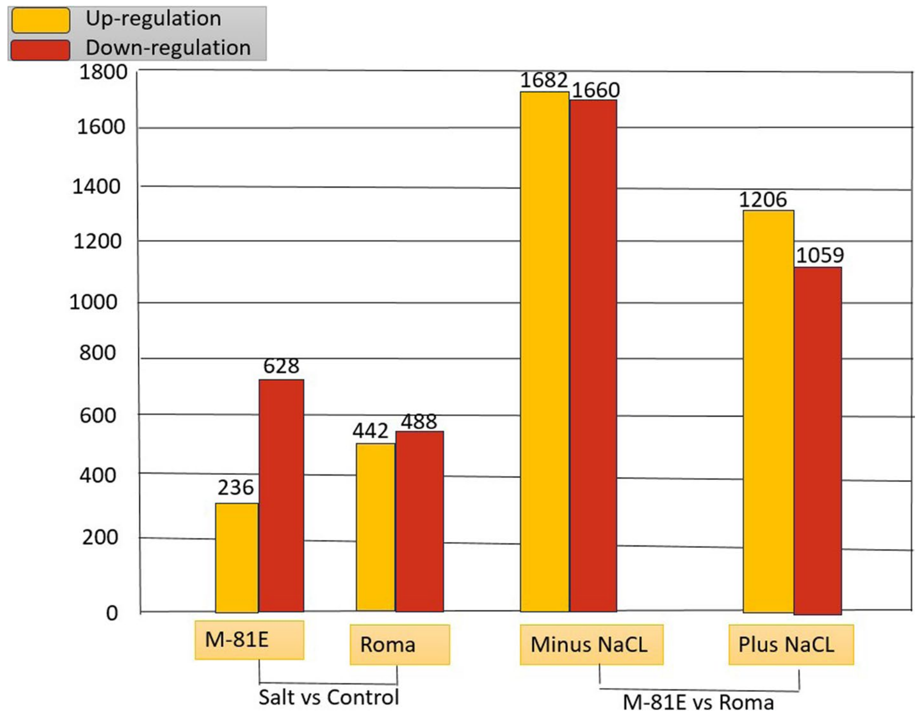

Fig. 5 The number of DEGs from various genotypes of Sorghum under salinity stress conditions (Sui et al. 2015)

were mapped to 878 distinct transcripts, of which 13 were associated with cellular components, 12 with molecular function, and 22 with biological processes. The category of binding had the most GO terms for molecular function, closely followed by the category of catalytic activity (Sui et al. 2015).

Functional classification of COG

Additionally, Sui et al. (2015) conducted Clusters of Orthologous Groups (COG) classification as a search criterion for all DEGs. The 864 DEGs contained 349 sequences that displayed an M-81E COG categorization. The largest cluster was found in the COG category for general function prediction only, which is followed by transcription, secondary metabolites biosynthesis, transport, and catabolism, amino acid transport and metabolism, carbohydrate transport and metabolism and amino acid transport and metabolism. The terms extracellular structure, nuclear structure, and chromatin structure and dynamics did not have any related genes. Cultivar Roma's COG classifications could be applied to 360 of the 930 sequences. The largest cluster was general function prediction only, followed by signal transduction mechanisms, transcription, and gene regulation, carbohydrate transport and metabolism, and replication, recombination, and repair. Unigene assignments were not made for extracellular structure, cell motility, nuclear structure, as well as intracellular trafficking, secretion, and vesicular transport.

KEGG's classification

Sweet sorghum transcriptome for putative biological pathways using the Kyoto Encyclopaedia of Genes and Genomes (KEGG) database was analyzed by Sui et al. (2015). A total of 70 and 63 KEGG pathways, respectively, received 150 DEGs of cultivar M-81E and 174 DEGs of cultivar Roma, respectively. They deduced that salt stress primarily affected photosynthesis and carbohydrate metabolism in sweet sorghum because the majority of these DEGs belong to the categories of photosynthesis, photosynthesis-antenna proteins, carbon fixation in photosynthetic organisms, and starch and sucrose metabolism.

Early in the process of photosynthesis, light energy is transformed into chemical energy. The outer light-harvesting complexes (LHCs) absorb a significant portion of the light which are tangentially coupled to PSI and PSII and contain the majority of chlorophyll and carotenoid pigments. Sui et al. (2015) mapped 8 DEGs from M-81E and 14 DEGs from sorghum cultivar Roma to the antenna proteins. Compared to the untreated control, both genotypes' expression of DEGs encoding Lhca1 and Lhcb1-5 was down-regulated by salt stress. However, after salt stress, DEGs encoding Lcha2-4 and Lchb6 showed decreased expression in cultivar Roma but remained unchanged in M-81E. The photosynthetic rate is strongly impacted by salt stress. The four proteins that make up the photosynthetic electron transport chain are in charge of moving electrons from water to $NADP^+$. These proteins include ATP synthase, the cytochrome (Cytb6f) complex, Photosystem II (PSII), and Photosystem I (PSI). In M-81E and Roma, 11 and 20 DEGs, respectively, changed the four protein components involved in the photosynthesis process in terms of both structure and function. Differentially Expressed Genes (DEGs) play a crucial role in understanding the molecular mechanisms underlying biological processes, including those involved in pathways like those found in the KEGG (Kyoto Encyclopedia of Genes and Genomes) database. The analysis of DEGs within KEGG pathways in sorghum provides valuable insights into the molecular processes underlying various biological phenomena.

Differentially expressed genes (DEGs) in response to drought stress

Drought is one of the most important environmental elements influencing agricultural productivity and dispersion. Plant stress symptoms associated with drought include a rapid suppression of root and, to a lesser extent, shoot growth. Leaf samples revealed 510, 559, and 3687 DEGs responding to treatments for mild drought, severe drought, and re-watering, respectively. In response to mild drought stress, 228 genes were up-regulated and 282 genes were down-regulated. In the genes responding to therapy for severe drought stress, 237 genes were down-regulated and 322 genes were up-regulated. There were 2472 up-regulated genes and 1215 down-regulated genes in leaves between a severe drought and rewatering treatment (Table 4) (Sui et al. 2015). One hundred thirty-four of the 1644 frequent DEGs found in roots, including abscisic acid (ABA), auxin (IAA), jasmonic acid (JA), ethylene (ET), and signal transduction

Table 4 DEGs (differentially expressed gene) under severe drought, mild drought and re-watering treatment in Sorghum (Source: Sui et al. 2015)

Sample	No. of DEGs	No. of upregulated genes	No. of down-regulated genes
Mild drought/control			
Roots	3368	1540	1828
Leaves	510	282	228
Severe drought/control			
Roots	5093	1750	3343
Leaves	559	322	237
Re-watering/severe drought			
Roots	4635	2825	1810
Leaves	3687	2472	1215

routes and metabolic activities of brassinosteroids (BR). One gene for zeaxanthin epoxidase (*ZEP, Sb06g018220*), one for beta-carotene 3-hydroxylase (*BCH, Sb01g048860*), and two for indole-3-acetaldehyde are part of drought response and adaptation pathways in sorghum.*Sb02g003230*, a different (9-*cis*-epoxycarotenoid dioxygenase) *NCED* (9-cis-epoxycarotenoid dioxygenase) gene, was up-regulated after severe drought treatments while down-regulated during re-watering treatments. Two ABA genes, *Sb02g026600* and *Sb04g030660*, which are involved in the oxidative catabolism of ABA, were likewise increased in response to treatments of mild and severe drought but downregulated in response to treatments of re-watering. In response to mild, severe, and rewatering drought treatments, the oxidase genes (*AO, Sb01g005650*, and *Sb02g003720*) were up-regulated in roots, suggesting that they may be involved in the conversion of abscisic aldehyde to abscisic acid (Shawrang et al. 2011; Sui et al. 2015).

Transcription factors in response to drought stress

Transcriptomic analysis is essential to understand the molecular responses of Sorghum to drought stress. It provides a comprehensive view of the genes that are differentially expressed under water scarcity, offering insights into the underlying mechanisms of stress adaptation. Only four transcription factors for heat stress were found among the 190 frequently occurring DEGs reacting to drought stress and re-watering therapy in leaves. Genes for ethylene-responsive transcription factor (ERF), heat shock transcription factor (HSF), N-terminal region (NAC) were discovered. A total of 66 DEGs encoding transcription were found in roots of the 1644 common DEGs that responded to mild, severe drought stress and re-watering treatments described above. HSF, ERF, NAC, WRKY, HD-ZIP (Homeodomain Leucine Zipper), bHLH (Basic Helix-Loop-Helix), and MYB (Myeloblastosis Viral Oncogene Homolog) factors were also discovered. Under mild and severe drought conditions, six HSF family genes and their downstream HSP genes were up-regulated, and under re-watering therapy, they were down-regulated. Under drought stress, two *HD-ZIP*, one *bHLH*, and one *MYB* gene were upregulated, under re-watering, they were down-regulated. One of the six *ERF* genes (*Sb01g049400*) was expressed in response to drought (Sui et al. 2015). The transcripts that code for the Late Embryogenesis Abundant (LEA) proteins are among the most highly increased transcripts. HKT1, a sodium ion transmembrane transporter essential in maintaining cellular Na^+ homeostasis, and P5CS2, which is involved in the metabolism of the compatible solute proline, are two other significantly affected genes (Fig. 6) (An et al. 2022).

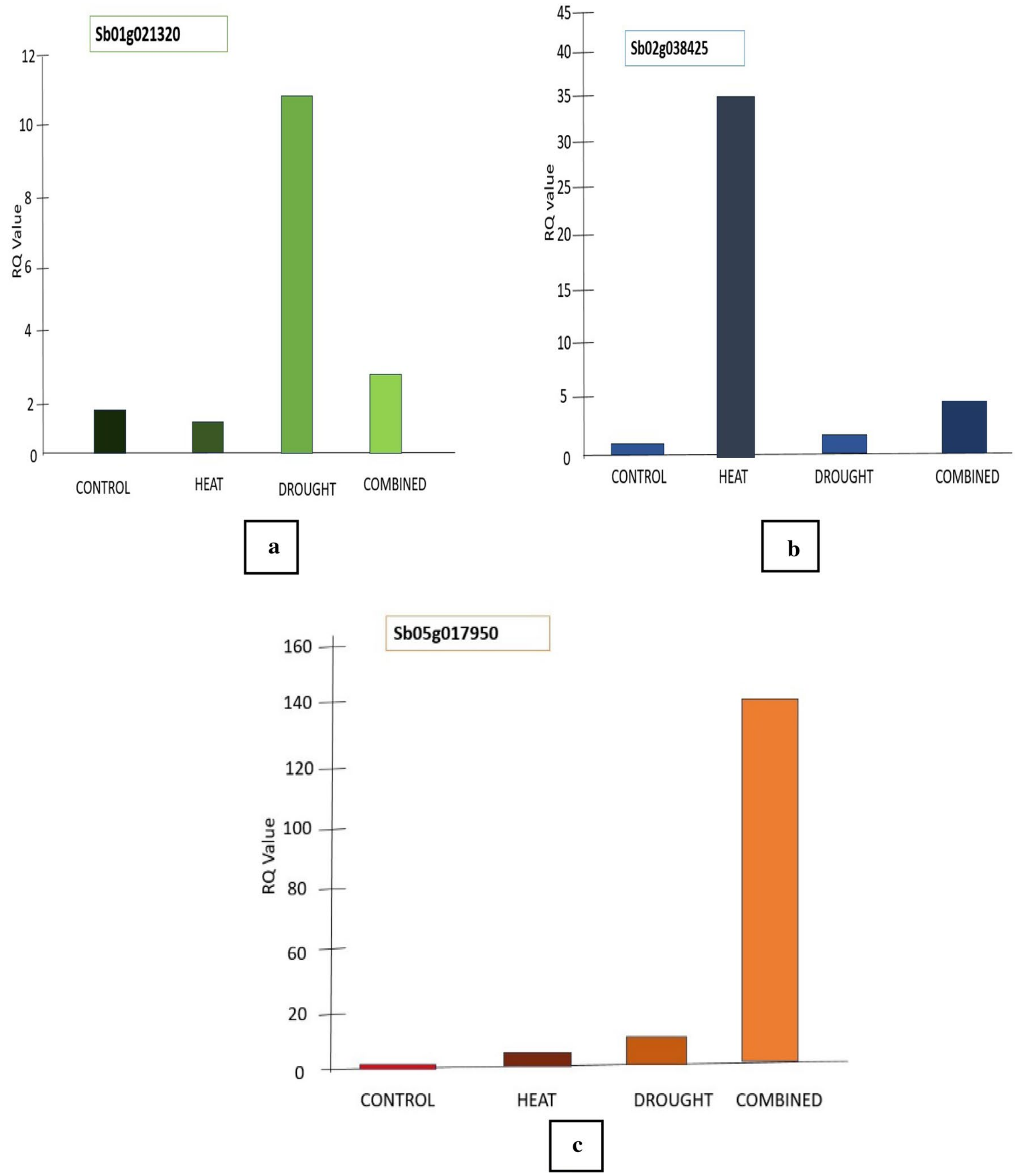

Fig. 6 Relative transcript abundance of genes from the gene sets identified as being selectively up-regulated by either drought stress (**a**), heat stress (**b**), or combined heat and drought stress (**c**). **a–c** depicts the effect occur during control, heat, drought and combined of different genes (An et al. 2022)

The stress caused by both heat and drought transcripts was up-regulated for 3003 and down-regulated for 2776 genes. A significant majority (60%) of these 5779 gene expression variations (both up and down) were shared with the sole response to heat stress, while 13% also responded to drought. None of the top 5 promoter motifs are associated with heat, despite this greater overlap were found to be enriched in genes that were elevated by heat and drought combined. In fact, the ABRE, (C/T) ACGTGTC, was the most comparable motif to the most enriched motif. There were 438 common gene expression changes across all 3 treatments (335 up-regulated and 103 down-regulated). Some of the particular genes in this collection that are connected to the general plant stress response are heat shock proteins, senescence-associated genes (SAGs), and glutathione transferases (Chen et al. 2016). Transcriptomics is crucial for understanding the genetic expression patterns in sorghum plants, providing insights into how genes are activated or repressed under different conditions. Transcriptomics is a powerful tool in sorghum plant research, providing a comprehensive understanding of gene expression patterns and molecular processes. The information derived from transcriptomic studies is instrumental in developing strategies for crop improvement, stress resilience, and the sustainable production of sorghum.

Future prospectus and economic importance

A limited amount of lysine and tryptophan is present in sorghum, but it contains both some essential and non-essential amino acids, such as alanine (7.34–9.62 g/100 g), aspartic acid (4.83–7.06 g/100 g), glutamic acid (17.5–28.12 g/100 g), leucine (12.02–14.48 g/100 g), and phenylalanine (4.03–5.62 g/100 g) (Oluwafemi 2020). Sorghum's steroid-binding abilities, involvement in reducing low-density lipoprotein (LDL), and role in preventing arthritis and rheumatism have all been observed. Flavonoids, condensed tannins, and phenolic acids (found only in a few cereal grains) are phenolic chemicals. The sorghum test and pericarp contain condensed tannins which protect the seed from insects, fungi, birds, and other rodents. The main goal of forage crop enhancement programs worldwide is biomass yield. Despite the growing difficulties in agriculture, there are still many opportunities, particularly in light of the rapidly developing omics technology. Owing to developments in biotechnology and computer sciences, omics data can now be produced reasonably priced for large collections of plants, species, or variants. The availability of cutting-edge computing and bioinformatics tools has made it possible to handle and analyze huge and complicated omics data. The identification of genes, their activity, types of RNA or proteins involved, their structures, and the processes involved in the development of the final morphological characteristic were made possible by the application of varied omics. Genomics aims to characterize and analyze the complete plant genome's genetic networks, compositions, organizations, and structures. When addressing complex features, genomic techniques are advantageous since these qualities frequently have a multigenic nature and a strong environmental influence. Transcriptomic technologies, which use high-throughput sequencing techniques to clarify coding and non-coding RNA expression levels in response to biotic and abiotic stresses on plants, give transcript data from RNA sequencing, microarray, and serial analysis of gene expression (SAGE).

Proteomics is the in-depth analysis of each protein in a complex biological system, such as the human body, plants, or animals during a particular period of time. To ascertain the relative abundance of various proteins, differences observed as a result of various post-translational modifications, and the ensuing function and localization of those proteins, proteomic analysis is required. It provides a brief summary of various metabolic processes, the relationships that result from them, and the effects they have on the regulatory pathways of other biological processes. Therefore, a proteome analysis is necessary to understand how distinct reactions of pathways behave under various strains and circumstances.

Metabolomics, an advanced biotechnological technique, addresses genotypes and phenotypic expressions of plants by identifying physiologically active metabolites, their functions, and the multiple biochemical processes in which they are involved. It is widely used to investigate various areas of plant breeding, including the selection of cultivars for agricultural usage, tolerance to biotic and abiotic challenges, nutritional progress, and the regulatory mechanisms driving plant growth and development. Metabolomes are basically the precursors and intermediates of the relevant biosynthetic pathways' metabolites, including secondary and primary, with low molecular weights (often around 1500 Da) (Kumar et al. 2022). Sorghum may be cultivated in a variety of soil conditions and is drought and waterlogging tolerant. Since it is a crop that can withstand water stress, semi-arid regions with little precipitation can support its cultivation. Africa is where it has its roots, and as such, it supports the Sorghum genotypes have the most genetic diversity.

Sorghum genetic modification (GM) shows itself as a practical and cost-effective substitute to enhance its nutritional characteristics or metabolism. However, a number of variables, such as regulatory frameworks, public opinion, and financial concerns, affect how feasible and acceptable genetically modified crops are. Gene modification can enhance nutritional content, biofortification, increased biotic and abiotic stresses. The acceptance of genetically modified agriculture is also greatly influenced by public opinion, moral issues, and concerns about the effects on the environment. The long-term implications and possible unexpected repercussions of genetic manipulation are also topics of continuing discussion. While genetic modification seems promising, its overall practicality and cost-effectiveness for enhancing the nutritional qualities or metabolism of sorghum must be determined through a thorough evaluation that takes into account ethical, environmental, and societal factors.

Sorghum plays a significant role in addressing food crises and contributing to food security due to its versatility, adaptability, and nutritional qualities. Sorghum is well-known for its resilience in arid and semi-arid regions, where other crops might struggle. Its ability to thrive in drought conditions makes it a reliable option in regions susceptible to water scarcity, contributing to stable crop yields and food production even during adverse climatic conditions. Sorghum is often grown by smallholder farmers in developing countries. Its cultivation supports rural livelihoods by providing income opportunities for farmers and contributing to local economies. The economic benefits derived from sorghum farming can help alleviate poverty and improve living standards in rural communities. Sorghum is used for biofuel production, contributing to the renewable energy sector. The economic benefits of sorghum-derived biofuels include energy security, and reduced dependence on non-renewable energy sources. Growing concerns about sustainable development and the need to meet the dual carbon targets of "peak carbon" and "carbon neutrality" have made it urgent to find effective ways to replace nonrenewable petroleum-based polymers with abundant and renewable plant-based components. This can be accomplished by creating biodegradable, mechanically pleasing, and reasonably priced biocomposites (Kan et al. 2023a, b). Sorghum's economic aspects make it a valuable crop in addressing food crises.

Conclusion

Combining omics technologies, such as genomics, proteomics, metabolomics and transcriptomics may facilitate the investigation of complex genetic and molecular pathways in cereal crops. Applying genome-wide scanning in conjunction with protein and metabolic profiling can efficiently identify desired agronomic traits, opening up new avenues for enhancing crop yields and resistances. A range of biotic and abiotic conditions can be investigated to determine the environmental effects of crops' phenotypic plasticity through high-throughput, phenomics-enabled genome-wide mapping, metabolic, and gene-expression studies (Kaur et al. 2021). The article highlights omics-based interventions in sorghum to combat food and nutritional scarcity in the future. The dynamic shifts in environmental conditions, including the impacts of global warming, pose challenges to plant production and may lead to potential food crises. However, directing our attention towards the cultivation of resilient crops adapted to challenging climates holds promise for significantly enhancing food productivity. This strategic focus on climate-resistant crops presents an opportunity to proactively address the potential adverse effects on agriculture and contribute to sustainable food security.

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Declarations

Conflict of interest All authors declare that they have no conflict of interest for writing and publication of this article.

References

Abdel-Ghany SE, Ullah F, Ben-Hur A, Reddy ASN (2020) Transcriptome analysis of drought-resistant and drought-sensitive sorghum (*Sorghum bicolor*) genotypes in response to peg-induced drought stress. Int J Mol Sci 21:1–26. https://doi.org/10.3390/ijms21030772

Abreha KB, Enyew M, Carlsson AS, Vetukuri RR, Feyissa T, Motlhaodi T, Ng'uniGeleta DM (2022) Sorghum in dryland: morphological, physiological, and molecular responses of sorghum under drought stress. Planta 255:1–23

Adebiyi AO, Adebiyi AP, Olaniyi EO (2005) Nutritional composition of sorghum bicolor starch hydrolyzed with amylase from *Rhizopus* sp. Afr J Biotechnol 4:1089–1094

Ahmad P, Bhardwaj R, Tuteja N (2012) Plant signalling under abiotic stress environment. In: Ahmad P, Prasad MNV (eds) Environmental adaptations and stress tolerance of plants in the era of climate change. Springer, New York, pp 297–323. https://doi.org/10.1007/978-1-4614-0815-4

Akpinar BA, Lucas SJ, Budak H (2013) Genomics approaches for crop improvement against abiotic stress. Sci World J. https://doi.org/10.1155/2013/361921

An HAM, Ahmed MF, Rashed MA (2022) Performance and transcriptomic analysis of *Sorghum bicolor* responding to drought stress. Sabrao J Breed Genet 54:814–825. https://doi.org/10.54910/sabrao2022.54.4.12

Ape DI, Nwogu NA, Uwakwe EI, Ikedinobi CS (2016) Comparative proximate analysis of maize and sorghum bought from ogbete main market of enugu state, Nigeria. Greener J Agric Sci 6:272–275. https://doi.org/10.15580/gjas.2016.9.101516167

Baillo EH, Hanif MS, Guo Y, Zhang Z, Xu P, Algam SA (2020) Genome-wide identification of WRKY transcription factor family members in sorghum (*Sorghum bicolor* (L.) moench). PLoS ONE 15(8):e0236651. https://doi.org/10.1371/journal.pone.0236651

Bueno PCP, Lopes NP (2020) Metabolomics to characterize adaptive and signaling responses in legume crops under abiotic stresses. ACS Omega 5:1752–1763. https://doi.org/10.1021/acsomega.9b03668

Cheatham CL, Sheppard KW (2017) Nutritional sciences. Cambridge Encycl Child Dev 793–798. https://doi.org/10.1017/9781316216491.126

Chen M, Rao RS, Zhang Y, Zhong C, Thelen JJ (2016) Metabolite variation in hybrid corn grain from a large-scale multisite study. Crop J 4(3):177–187. https://doi.org/10.1016/j.cj.2016.03.004

Chen D, Wang W, Wu Y, Xie H, Zhao L, Zeng Q, Zhan Y (2019) Expression and distribution of the auxin response factors in *Sorghum bicolor* during development and temperature stress. Int J Mol Sci 20(19):4816. https://doi.org/10.3390/ijms20194816

Ciarmiello LF, Woodrow P, Fuggi A, Pontecorvo G, Carillo P (2011) Plant genes for abiotic stress. Abiotic stress in plants–mechanisms and adaptations, vol 22. IntechOpen, London, pp 283–308

Cui J, Shen N, Lu Z, Xu G, Wang Y, Jin B (2020) Analysis and comprehensive comparison of PacBio and nanopore-based RNA sequencing of the Arabidopsis transcriptome. Plant Met 16:1–3. https://doi.org/10.1186/s13007-020-00629-x

Cui H, Chen J, Liu M, Zhang H, Zhang S, Liu D, Chen S (2022) Genome-wide analysis of C_2H_2 zinc finger gene family and its response to cold and drought stress in sorghum [*Sorghum bicolor* (L.) moench]. Int J Mol Sci 23(10):5571. https://doi.org/10.3390/ijms23105571

Fadoul HE, El Siddig MA, Abdalla AW, El Hussein AA (2018) Physiological and proteomic analysis of two contrasting *Sorghum bicolor* genotypes in response to drought stress. Aust J Crop Sci 12(9):1543–1551

Fernandez O, Urrutia M, Bernillon S, Giauffret C, Tardieu F, Le Gouis J, Langlade N, Charcosset A, Moing A, Gibon Y (2016) Fortune telling: metabolic markers of plant performance. Metabolomics 12:1–4. https://doi.org/10.1007/s11306-016-1099-1

Garcia-Caparros P, De Filippis L, Gul A, Hasanuzzaman M, Ozturk M, Altay V, Lao MT (2021) Oxidative stress and antioxidant metabolism under adverse environmental conditions: a review. Bot Rev 87:421–466. https://doi.org/10.1007/s12229-020-09231-1

Goche T, Shargie NG, Cummins I, Brown AP, Chivasa S, Ngara R (2020) Comparative physiological and root proteome analyses of two sorghum varieties responding to water limitation. Sci Rep 10(1):11835. https://doi.org/10.1038/s41598-020-68735-3

Gregory PJ, Ingram JSI, Brklacich M (2005) Climate change and food security. Philos Trans R Soc B Biol Sci 360:2139–2148. https://doi.org/10.1098/rstb.2005.1745

Grzybowski MW, Zwiener M, Jin H, Wijewardane NK, Atefi A, Naldrett MJ, Alvarez S, Ge Y, Schnable JC (2022) Variation in morpho-physiological and metabolic responses to low nitrogen stress across the sorghum association panel. BMC Plant Biol 22(1):1–4. https://doi.org/10.1186/s12870-022-03823-2

Gupta S, Arya GC, Malviya N, Bisht NC, Yadav D (2016) Molecular cloning and expression profiling of multiple Dof genes of *Sorghum bicolor* (L.) Moench. Mol Biol Rep 43:767–774. https://doi.org/10.1007/s11033-016-4019-6

Hao H, Li Z, Leng C, Lu C, Luo H, Liu Y, Wu X, Liu Z, Shang L, Jing HC (2021) Sorghum breeding in the genomic era: opportunities and challenges. Theor Appl Genet 134:1899–1924

Jagadish SVK, Way DA, Sharkey TD (2021) Plant heat stress: concepts directing future research. Plant Cell Environ 44:1992–2005. https://doi.org/10.1111/pce.14050

Jayakodi M, Schreiber M, Stein N, Mascher M (2021) Building pangenome infrastructures for crop plants and their use in association genetics. DNA Res 28(1):dsaa030. https://doi.org/10.1093/dnares/dsaa030

Jedmowski C, Ashoub A, Beckhaus T, Berberich T, Karas M, Brüggemann W (2014) Comparative analysis of *Sorghum bicolor* proteome in response to drought stress and following recovery. Int J Proteom. https://doi.org/10.1155/2014/395905

Jones RW, Beckwith AC (1970) Proximate composition and proteins of three grain sorghum hybrids and their dry-mill fractions. J Agric Food Chem 18:33–36. https://doi.org/10.1021/jf60167a002

Kan Y, Kan H, Bai Y, Zhang S, Gao Z (2023a) Effective and environmentally safe self-antimildew strategy to simultaneously improve the mildew and water resistances of soybean flour-based adhesives. J Clean Prod 392:136319

Kan Y, Li J, Zhang S, Gao Z (2023b) Novel bridge assistance strategy for tailoring crosslinking networks within soybean-meal-based biocomposites to balance mechanical and biodegradation properties. Chem Eng J 472:144858

Kaur G, Asthir B (2017) Molecular responses to drought stress in plants. Biol Plant 61:201–209. https://doi.org/10.1007/s10535-016-0700-9

Kaur B, Sandhu KS, Kamal R, Kaur K, Singh J, Röder MS, Muqaddasi QH (2021) Omics for the improvement of abiotic, biotic, and agronomic traits in major cereal crops: applications, challenges, and prospects. Plants 10(10):1989. https://doi.org/10.3390/plants10101989

Kim JY, Kim WY, Kwak KJ, Oh SH, Han YS, Kang H (2010) Glycine-rich RNA-binding proteins are functionally conserved in *Arabidopsis thaliana* and *Oryza sativa* during cold adaptation process. J Exp Bot 61(9):2317–2325. https://doi.org/10.1093/jxb/erq058

Kim SG, Lee JS, Kim JT, Kwon YS, Bae DW, Bae HH, Son BY, Baek SB, Kwon YU, Woo MO, Shin S (2015) Physiological and proteomic analysis of the response to drought stress in an inbred Korean maize line. Plant Omics. https://doi.org/10.21475/ajcs.18.12.09

Kosova K, Vitamvas P, Urban MO, Prasil IT, Renaut J (2018) Plant abiotic stress proteomics: the major factors determining alterations in cellular proteome. Front Plant Sci 9:122. https://doi.org/10.3389/fpls.2018.00122

Kumar P, Singh J, Kaur G, Adunola PM, Biswas A, Bazzer S, Kaur H, Kaur I, Kaur H, Sandhu KS, Vemula S (2022) OMICS in fodder crops: applications, challenges, and prospects. Curr Issues Mol Biol 44(11):5440–5473. https://doi.org/10.3390/cimb44110369

Labuschagne MT (2018) A review of cereal grain proteomics and its potential for sorghum improvement. J Cereal Sci 84:151–158. https://doi.org/10.1016/j.jcs.2018.10.010

Lee S, Choi YM, Shin MJ, Yoon H, Wang X, Lee Y, Desta KT (2023) Exploring the potentials of sorghum genotypes: a comprehensive study on nutritional qualities, functional metabolites, and antioxidant capacities. Front Nutr. https://doi.org/10.3389/fnut.2023.1238729

Li C, Lin F, An D, Wang W, Huang R (2017) Genome sequencing and assembly by long reads in plants. Genes 9(1):6. https://doi.org/10.3390/genes9010006

Li H, Li Y, Ke Q, Kwak SS, Zhang S, Deng X (2020) Physiological and differential proteomic analyses of imitation drought stress response in *Sorghum bicolor* root at the seedling stage. Int J Mol Sci 21(23):9174. https://doi.org/10.3390/ijms21239174

Li J, Bai X, Ran F, Zhang C, Yan Y, Li P, Chen H (2024) Effects of combined extreme cold and drought stress on growth, photosynthesis, and physiological characteristics of cool-season grasses. Sci Rep 14(1):116

Lisec J, Römisch-Margl L, Nikoloski Z, Piepho HP, Giavalisco P, Selbig J, Gierl A, Willmitzer L (2011) Corn hybrids display lower metabolite variability and complex metabolite inheritance patterns. Plant J 68(2):326–336

Litvinov DY, Karlov GI, Divashuk MG (2021) Metabolomics for crop breeding: general considerations. Genes (basel). https://doi.org/10.3390/genes12101602

Liu M, Liu S, Yu X, Zhang L, Bao L, Chai X, Zhang L, Wang R, Lu P, Guo G (2024) Understanding the salt stress response of *Sorghum bicolor* (L.): a full-length transcriptome analysis via ONT sequencing. https://doi.org/10.2139/ssrn.4684598

Loskutov IG, Shelenga TV, Konarev AV (2020) Modern approach of structuring the variety diversity of the naked and covered forms of cultural oats (*Avena sativa* L.). Ecol Genet 18:27–41. https://doi.org/10.17816/ecogen12977

Lu M, Zhang DF, Shi YS, Song YC, Wang TY, Li Y (2013) Expression of SbSNAC1, a NAC transcription factor from sorghum, confers drought tolerance to transgenic Arabidopsis. Plant Cell Tissue Organ Cult 115:443–455. https://doi.org/10.1007/s11240-013-0375-2

Mareya CR, Tugizimana F, Piater LA, Madala NE, Steenkamp PA, Dubery IA (2019) Untargeted metabolomics reveal defensome-related metabolic reprogramming in *Sorghum bicolor* against infection by *Burkholderia andropogonis*. Metabolites 9(1):8

McCormick RF, Truong SK, Sreedasyam A, Jenkins J, Shu S, Sims D, Kennedy M, Amirebrahimi M, Weers B, McKinley B, Mattison A (2018) The *Sorghum bicolor* reference genome: improved assembly and annotations, a transcriptome atlas, and signatures of genome organization. https://doi.org/10.1101/110593

Nagaraju M, Sudhakar Reddy P, Anil Kumar S, Srivastava RK, Kavi Kishor PB, Rao DM (2015) Genome-wide scanning and characterization of *Sorghum bicolor* L. heat shock transcription factors. Curr Genom 16(4):279–291

Nayak SN, Aravind B, Malavalli SS, Sukanth BS, Poornima R, Bharati P, Hefferon K, Kole C, Puppala N (2021) Omics technologies to enhance plant based functional foods: an overview. Front Genet 12:742095

Ndimba RJ, Kruger J, Kossmann J, Ndimba BK (2017) A comparative study of selected physical and biochemical traits of wild-type and transgenic sorghum to reveal differences relevant to grain quality. Front Plant Sci 8:247680

Neilson KA, Gammulla CG, Mirzaei M, Imin N, Haynes PA (2010) Proteomic analysis of temperature stress in plants. Proteomics 10(4):828–845

Ngara R, Goche T, Swanevelder DZH, Chivasa S (2021) Sorghum's whole-plant transcriptome and proteome responses to drought stress: a review. Life. https://doi.org/10.3390/life11070704

Ogden AJ, Abdali S, Engbrecht KM, Zhou M, Handakumbura PP (2020) Distinct pre-flowering drought tolerance strategies of *Sorghum bicolor* genotype RTx430 revealed by subcellular protein profiling. Int J Mol Sci 21(24):9706

Okoh PN, Obilana AT, Njoku PC, Aduku AO (1982) Proximate analysis, amino acid composition and tannin content of improved nigerian sorghum varieties and their potential in poultry feeds. Anim Feed Sci Technol 7:359–364. https://doi.org/10.1016/0377-8401(82)90005-0

Oluwafemi AA (2020) African sorghum-based fermented foods: past, current and future prospects. Nutrients 12:1111–1136

Paterson AH, Bowers JE, Bruggmann R, Dubchak I, Grimwood J, Gundlach H, Haberer G, Hellsten U, Mitros T, Poliakov A, Schmutz J (2009) The *Sorghum bicolor* genome and the diversification of grasses. Nature 457(7229):551–556

Piveta LB, Roma-Burgos N, Noldin JA, Viana VE, Oliveira CD, Lamego FP, Avila LA (2020) Molecular and physiological responses of rice and weedy rice to heat and drought stress. Agriculture 11(1):9. https://doi.org/10.3390/agriculture11010009

Prasad VR, Govindaraj M, Djanaguiraman M, Djalovic I, Shailani A, Rawat N, Singla-Pareek SL, Pareek A, Prasad PV (2021) Drought and high temperature stress in sorghum: physiological, genetic, and molecular insights and breeding approaches. Int J Mol Sci 22(18):9826

Proveniers MCG, Van Zanten M (2013) High temperature acclimation through PIF4 signaling. Trends Plant Sci 18:59–64. https://doi.org/10.1016/j.tplants.2012.09.002

Quirino BF, Candido ES, Campos PF, Franco OL, Krüger RH (2010) Proteomic approaches to study plant–pathogen interactions. Phytochemistry 71(4):351–362

Ramalingam A, Kudapa H, Pazhamala LT, Weckwerth W, Varshney RK (2015) Proteomics and metabolomics: two emerging areas for legume improvement. Front Plant Sci 6:1116. https://doi.org/10.3389/fpls.2015.01116

Razzaq A, Sadia B, Raza A, Khalid Hameed M, Saleem F (2019) Metabolomics: a way forward for crop improvement. Metabolites 9(12):303. https://doi.org/10.3390/metabo9120303

Riedelsheimer C, Czedik-Eysenberg A, Grieder C, Lisec J, Technow F, Sulpice R, Altmann T, Stitt M, Willmitzer L, Melchinger AE (2012) Genomic and metabolic prediction of complex heterotic traits in hybrid maize. Nat Genet 44(2):217–220

Rodriguez Cubillos AE, Tong H, Alseekh S, de Abreu e Lima F, Yu J, Fernie AR, Nikoloski Z, Laitinen RA (2018) Inheritance patterns in metabolism and growth in diallel crosses of *Arabidopsis thaliana* from a single growth habitat. Heredity 120(5):463–473. https://doi.org/10.1038/s41437-017-0030-5

Roy SJ, Tucker EJ, Tester M (2011) Genetic analysis of abiotic stress tolerance in crops. Curr Opin Plant Biol 14(3):232–239. https://doi.org/10.1016/j.pbi.2011.03.002

Roychoudhury A, Paul S, Basu S (2013) Cross-talk between abscisic acid-dependent and abscisic acid-independent pathways during abiotic stress. Plant Cell Rep 32:985–1006. https://doi.org/10.1007/s00299-013-1414-5

Sah SK, Reddy KR, Li J (2016) Abscisic acid and abiotic stress tolerance in crop plants. Front Plant Sci 7:571. https://doi.org/10.3389/fpls.2016.00571

Sanjari S, Shirzadian-Khorramabad R, Shobbar ZS, Shahbazi M (2019) Systematic analysis of NAC transcription factors' gene family and identification of post-flowering drought stress responsive members in sorghum. Plant Cell Rep 38:361–376. https://doi.org/10.1007/s00299-019-02371-8

Sarkar MAR, Sarkar S, Islam MSU, Zohra FT, Rahman SM (2023) A genome-wide approach to the systematic and comprehensive analysis of LIM gene family in sorghum (*Sorghum bicolor* L.). Genom Inform 21(3):e36

Scossa F, Alseekh S, Fernie AR (2021) Integrating multi-omics data for crop improvement. J Plant Physiol 257:153352. https://doi.org/10.1016/j.jplph.2020.153352

Shanker AK, Maddaala A, Kumar MA, Yadav SK, Maheswari M, Venkateswarlu B (2012) *In silico* targeted genome mining and comparative modelling reveals a putative protein similar to an Arabidopsis drought tolerance DNA binding transcription factor in chromosome 6 of *Sorghum bicolor* genome. Interdiscip Sci Comput Life Sci 4:133–141

Sharma S, Singh E, Jain P (2015) Effect of different household processing on nutritional and anti-nutritional factors in *Vigna aconitifolia* and *Sorghum bicolour* (L.) moench seeds and their product development. J Med Nutr Nutraceuticals 4:95. https://doi.org/10.4103/2278-019x.151809

Shawrang P, Sadeghi AA, Behgar M, Zareshahi H, Shahhoseini G (2011) Study of chemical compositions, anti-nutritional contents and digestibility of electron beam irradiated sorghum grains. Food Chem 125(2):376–379

Sheflin AM, Chiniquy D, Yuan C, Goren E, Kumar I, Braud M, Brutnell T, Eveland AL, Tringe S, Liu P, Kresovich S (2019) Metabolomics of sorghum roots during nitrogen stress reveals compromised metabolic capacity for salicylic acid biosynthesis. Plant Direct 3(3):e00122

Song EH, Jeong J, Park CY, Kim HY, Kim EH, Bang E, Hong YS (2018) Metabotyping of rice (*Oryza sativa* L.) for understanding its intrinsic physiology and potential eating quality. Food Res Int 111:20–30

Steinfath M, Strehmel N, Peters R, Schauer N, Groth D, Hummel J, Steup M, Selbig J, Kopka J, Geigenberger P, Van Dongen JT (2010) Discovering plant metabolic biomarkers for phenotype prediction using an untargeted approach. Plant Biotechnol J 8(8):900–911

Sui N, Yang Z, Liu M, Wang B (2015) Identification and transcriptomic profiling of genes involved in increasing sugar content during salt stress in sweet sorghum leaves. BMC Genom 16:1–18. https://doi.org/10.1186/s12864-015-1760-5

Tamhane VA, Sant SS, Jadhav AR, War AR, Sharma HC, Jaleel A, Kashikar AS (2021) Label-free quantitative proteomics of *Sorghum bicolor* reveals the proteins strengthening plant defense against insect pest *Chilo partellus*. Proteome Sci 19(1):1–25. https://doi.org/10.1186/s12953-021-00173-z

Tan BC, Lim YS, Lau SE (2017) Proteomics in commercial crops: an overview. J Proteom 169:176–188. https://doi.org/10.1016/j.jprot.2017.05.018

Teferra TF (2019) Quinoa and other andean ancient grains: super grains for the future. Cereal Foods World. https://doi.org/10.1094/cfw-64-5-0053

Thomason K, Babar MA, Erickson JE, Mulvaney M, Beecher C, MacDonald G (2018) Comparative physiological and metabolomics analysis of wheat (*Triticum aestivum* L.) following post-anthesis heat stress. PLoS ONE 13(6):e0197919

Tian T, You Q, Zhang L, Yi X, Yan H, Xu W, Su Z (2016) SorghumFDB: sorghum functional genomics database with multidimensional network analysis. Database 1:baw099

Tu M, Du C, Yu B, Wang G, Deng Y, Wang Y, Li Y (2023) Current advances in the molecular regulation of abiotic stress tolerance in sorghum via transcriptomic, proteomic, and metabolomic approaches. Front Plant Sci 14:1147328

Udachan IS, Sahoo AK, Hend GM (2012) Extraction and characterization of sorghum (Sorghum bicolor L. moench) starch. Int Food Res J 19:315–319

Wang S, Bai Y, Shen C, Wu Y, Zhang S, Jiang D, Guilfoyle TJ, Chen M, Qi Y (2010) Auxin-related gene families in abiotic stress response in *Sorghum bicolor*. Funct Integr Genom 10:533–5546

Wang X, Wang T, Xu J, Shen Z, Yang Y, Chen A, Wang S, Liang E, Piao S (2022) Enhanced habitat loss of the Himalayan endemic flora driven by warming-forced upslope tree expansion. Nature Ecol Evol 6(7):890–909

Woldesemayat AA, Modise DM, Ndimba BK (2018) Identification of proteins in response to terminal drought stress in sorghum (*Sorghum bicolor* (L.) Moench) using two-dimensional gel-electrophoresis and MALDI-TOF-TOF MS/MS. Indian J Plant Physiol 23:24–39

Wu X, Wang W (2016) Increasing confidence of proteomics data regarding the identification of stress-responsive proteins in crop plants. Front Plant Sci 7:702. https://doi.org/10.3389/fpls.2016.00702

Wu XY, Hu WJ, Luo H, Xia Y, Zhao Y, Wang LD, Zhang LM, Luo JC, Jing HC (2016) Transcriptome profiling of developmental leaf senescence in sorghum (*Sorghum bicolor*). Plant Mol Biol 92:555–580. https://doi.org/10.1007/s11103-016-0532-1

Xue Y, Bai X, Zhao C, Tan Q, Li Y, Luo G, Long M, Wu L, Chen F, Li C, Ran C, Zhang S, Liu M, Gong S, Xiong L, Song F, Du C, Xiao B, Li Z, Long M (2023) Spring photosynthetic phenology of Chinese vegetation in response to climate change and its impact on net primary productivity. Agric for Meteorol 342:109734

Yang Z, Chi X, Guo F, Jin X, Luo H, Hawar A, Chen Y, Feng K, Wang B, Qi J, Yang Y (2020) SbWRKY30 enhances the drought tolerance of plants and regulates a drought stress-responsive gene, SbRD19, in sorghum. J Plant Physiol 246:153142

Zenda T, Liu S, Dong A, Li J, Wang Y, Liu X, Wang N, Duan H (2021) Omics-facilitated crop improvement for climate resilience and superior nutritive value. Front Plant Sci 12:774994. https://doi.org/10.3389/fpls.2021.774994

Zhang C, Hao YJ (2020) Advances in genomic, transcriptomic, and metabolomic analyses of fruit quality in fruit crops. Hortic Plant J 16(6):361–371. https://doi.org/10.1016/j.hpj.2020.11.001

Zhang HH, Xu N, Wu X, Wang J, Ma S, Li X, Sun G (2018) Effects of four types of sodium salt stress on plant growth and photosynthetic apparatus in sorghum leaves. J Plant Int 13(1):506–513

Zhou Y, Wang Z, Li Y, Li Z, Liu H, Zhou W (2020) Metabolite profiling of sorghum seeds of different colors from different sweet sorghum cultivars using a widely targeted metabolomics approach. Int J Genom.https://doi.org/10.1155/2020/6247429

Zhang X, Zong J, Liu J, Yin J, Zhang D (2010) Genome-wide analysis of WOX gene family in rice, sorghum, maize, Arabidopsis and poplar. J Integr Plant Biol 52(11):1016–1026